NASA SDS Product Specification

Level-2 Geocoded Single Look Complex

L2_GSLC

Rev E

JPL D-102269

Nov 08, 2024, Version 1.2.1

Author: Heresh Fattahi

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National Aeronautics and
Space Administration

Jet Propulsion Laboratory
California Institute of Technology
Pasadena, California

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SIGNATURE PAGE

Prepared by:

Electronic signature on file _____ *07-Jan-2025*_____
Heresh Fattahi, NISAR ADT Lead Date

Approved by:

Electronic signature on file _____ *03-Jan-2025*_____
Ana Maria Guerrero, NISAR Mission System Manager Date

Electronic signature on file _____ *02-Jan-2025*_____
Paul Rosen, NISAR Project Scientist Date

Electronic signature on file _____ *06-Jan-2025*_____
Cecilia Cheng, NISAR SDS Manager Date

Electronic signature on file _____ *06-Jan-2025*_____
Helen Mortensen, NISAR SDS Lead System Engineer Date

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EPDM ELECTRONIC SIGNATURES

User-Group/Role	...	Decision	...	Date
Cheng, Cecilia S (cecilia)-JPL Consumer/Project C...	...	Approve		06-Jan-2025 15:42
Fattahi, Heresh (fattahi)-JPL Author/JPL Author WIP	...	Approve		07-Jan-2025 11:21
Guerrero, Ana Maria P (ana)-JPL Author/JPL Author	...	Approve		03-Jan-2025 08:10
Mortensen, Helen B (hbmorten)-JPL Consumer/Pr...	...	Approve		06-Jan-2025 09:03
Rosen, Paul A (parosen)-JPL Consumer/Project C...	...	Approve		02-Jan-2025 19:29

ACKNOWLEDGEMENT

The research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004).

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DOCUMENT CHANGE LOG

Revision	Cover Date	Sections Changed	ECR #	Reason, ECR Title, LRS #*
Preliminary	Sep 5, 2019	All	N/A	New document LRR044747 for ISRO LRR044748 for JPL Sharma LRR044749 for JPL Brancato and Shiroma
Rev A	June 16, 2020	Sec 1.3. Sec 2.1, 2.2. Sec 5.2, 5.3, 5.5		Release 1. Update document lists. Update product flow. Update product content.
Rev A ER R1.1	December 7, 2020	None	N/A	Minor wording cleanup. No product changes. For completeness for ER1.1
Rev A (R1.1) Working Version	February, 5, 2021	Sec 2.1, Sec 3.2 , Sec 4.4		For Release R1.1. LRR-054210 for NISAR Science Team
Rev A (R2.0) Working Version 2	May 13, 2021	Sec 1.2, Sec 2.1, Sec 2.2, Sec 6		Merged info from Product Description Document (PDD). Removed reference to PDD. Added Section 6 on metadata cubes. LRR056761 for NISAR Science Team LRR056768 for ISRO
Rev A (R2.0) Working Version	July 15, 2021	Sec 1,2,3,4		Minor wording cleanup. No product changes.
Rev A (ER 3.0) Working Version	December 10, 2021	Sec 3.2.3		For Engineering Release ER 3.0.
Rev A (R3) Working Version	March 04, 2022	Sec.1.3, Figure 2-1, Table 2-1, Appendix A		Updated Docushare link in Sec. 1.3. Added description of offset products in Table 2-1. Updated product dependency Figure 2-1. Added missing acronyms in Appendix A. Removed incomplete LRR number in the footer.
Rev A (R3.1) Working Version	August 05, 2022	Signature page, Sec 1.3	N/A	Updated signature page; updated reference to the RSLC Product Specification Document; Added offset products to Figure 2-1.
Rev A (R3.3)	June 6, 2023	Signature page, sec 2.2, 3.2, 4.4.4, App A	N/A	For R3.3 URS CL# 23-3837
Rev B (R3.4)	Oct 23, 2023	Sec 5		Added projection metadata to different datasets for consistency with CF convention
Rev C (R4.0)	February 07, 2024	Cover page, Sec 1.3, Figure2-1, Table-2-1, Table 2-2, Table 2-3; Sec 5		Added version number to cover page; Updated AD and RD; Updated Fig. 2-1; Added Sec 2.2.4 on geo referencing the products, Updated Table 2-3 and clarified caption. Sec 2.2 with additional information on product spacing. Sec 5 all tables updated with latest product xml. Cleared for public release. URS CL# 24-1204.
Rev D (R4.0.2)	May 31, 2024	Cover page, Sec 3.8, Sec 4.2.1, Sec 4.4, Sec 5.1, 5.4, 5.6		Updated version number and date on cover page, added cloud optimized HDF5 subsection 3.8, Added CRID to sec 4.1, Added mask layer definition to Sec

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				4.4. Table 5-1 added dopplerCentroidTimeLength and dopplerCentroidSlantRangeWidth, numberOfInputTecFiles, Sec. 5.2: Added or updated boundingPolygon, compositeReleaseId. Sec. 5.4: Added parameter rfiLikelihood to all frequencies and polarizations; deleted crosstalk/projection; updated Crosspol. Sec. 5.5: Added tecFiles. Sec. 5.6. Added or updated interpMethod, orbitType, attitudeType. Cleared for public release. URS CL#24-3353.
Rev E (R4.0.4)	Nov 8, 2024	Cover page, Sec 1.3, Sec 3.1.4, Sec 3.2.3, Sec 3.3, Sec 3.4, Sec 3.6, Sec 3.7, Sec 4.5.1, Sec 4.5.2.1, Sec 4.5.2.2, Tables 5-1 through 5-7		Updated version number and date on cover page. Updated reference documents. Clarified data type, variable metadata, granule, and “string” definitions to address feedback from DAAC. Added current product file naming convention. Added information about output grid. Added time convention. Added metadata for CEOS compliance. Added grid_mapping_name, longitude_of_central_meridian, and scale_factor_at_central_meridian attributes for CF-1.7 compliance. Added Digital Object Identifier (DOI) to product metadata as per ESDIS guidelines. URS CL#24-6471.
* *Include the JPL Limited Release System (LRS) clearance number for each revision to be shared with foreign partners.*				

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1 INTRODUCTION

1.1 Purpose of Description

This document provides a specification of the NASA-ISRO Synthetic Aperture Radar (NISAR) L-SAR Level-2 Geocoded Single Look Complex product to be generated by the NASA Science Data System (SDS) and provided to the Distributed Active Archive Center (DAAC). This data product is referenced by the short name GSLC.

1.2 Document Organization

Section 2 provides an overview of the product.

Section 3 provides the structure of the product, including granule definition, file organization, spatial resolution, temporal and spatial organization of the content.

Section 4 provides qualitative descriptions of the information provided in the product.

Section 5 provides a detailed identification of the individual fields within the GSLC product, including for example their units, size, coordinates, etc.

Section 6 provides a description of the metadata cube representation.

Appendix A provides a listing of the acronyms used in this document.

Appendix B provides a description of geolocation grids and projection systems used for the product.

1.3 Applicable and Reference Documents

Applicable documents levy requirements on areas addressed in this document. Reference documents are cited to provide additional information to readers. In case of conflict between the applicable documents and this document, the Project shall review the conflict to find the most effective resolution.

Applicable Documents

[AD1] NISAR NASA SDS Level 4 Requirements, JPL D-95655, Rev A, February 06, 2024

[AD2] NISAR NASA SDS Algorithm Development Plan, JPL D-95678, Initial, Sep. 12, 2019

[AD3] NISAR Science Data Management and Archive Plan, JPL D-80828, June 1, 2016

[AD4] NISAR Science Management Plan, JPL D-76340, Rev A, Aug. 14, 2018

[AD5] NISAR SDS ADT Calibration and Validation Plan, JPL D-102256, Rev A, November 20, 2023

[AD6] NISAR NASA SDS L4 Software Management Plan (SMP), JPL D-95656, Rev A, Sep. 19, 2022

[AD7] ISO-19115-2, https://www.iso.org/obp/ui/#iso:std:iso:19115:-2:ed-2:v1:en

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Reference Documents

[RD1] NISAR NASA SDS Algorithm Theoretical Basis Document, JPL D-95677, Rev A, November 12, 2023.

[RD2] EOSDIS Handbook, July 2016, retrieved from https://cdn.earthdata.nasa.gov/conduit/upload/5980/EOSDISHandbookWebFinaL2.pdf

[RD3] NISAR SDS File Naming Conventions, JPL D-102255, Initial, Nov. 4, 2020

[RD4] NISAR L1_RSLC Product Specification Document, JPL D-102268,

[RD5] HDF5 documentation at https://portal.hdfgroup.org/documentation, Accessed May 30, 2024.

[RD6] Eineder, M. (2003), Efficient simulation of SAR interferograms of large areas and of rugged terrain, IEEE Transactions on Geoscience and Remote Sensing, 41(6), 1415-1427.

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2 PRODUCT OVERVIEW

2.1 Product Background

Each NASA SDS L0-L2 LSAR product (
Figure 2-1 and Table 2-1 Product Dependency) is distributed as a single Hierarchical Data Format version 5 (HDF5, [RD5]) granule. All the metadata and imagery data are packaged in clearly defined sub-groups within the granule in compliance with the HDF5 specification. The NISAR product level definitions are given in Table 2-2.

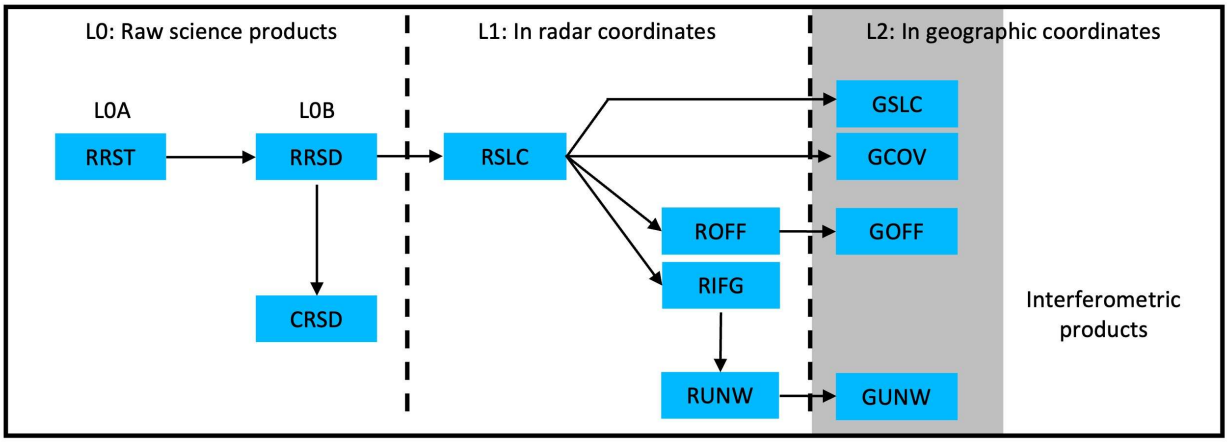

Figure 2-1 Product Dependency

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Table 2-1. Key to Product Dependency Diagram

L0 Product	Scope	Description	Granule Size
Radar Raw Science Telemetry (RRST)	Global	This L0A product contains the raw downlinked data delivered to SDS	By downlinked files
Radar Raw Signal Data (RRSD)	Global	This L0B product is corrected, aligned radar pulse data derived from the RRST products and used for further processing	By radar observation, i.e., continuous data collected in a single radar mode
Calibration Raw Signal Data (CRSD)	Global	This L0B product contains instrument calibration data.	By radar datatake, i.e., a sequence of observations for one radar-on period

L1 Product	Scope	Description	Granule Size
Range-Doppler Single Look Complex (RSLC)	Global	The L1 RSLC product contains focused SAR images in range-Doppler coordinates. The RSLC is input to other L1 or L2 products.	On pre-defined track/frame.
Range-Doppler Nearest-Time Interferogram (RIFG)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only.	Multi-looked interferogram in Range Doppler coordinates, ellipsoid and topographic phase flattened and formed with precise coregistration using geometrical offsets and high-resolution pixel offsets obtained from incoherent cross correlation.	On pre-defined track/frame
Range-Doppler Nearest-Time Pixel Offsets (ROFF)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only.	Unfiltered and unculled layers of pixel offsets in Range Doppler coordinates with different resolutions obtained from incoherent cross correlation.	On pre-defined track/frame
Range-Doppler Nearest-Time Unwrapped Interferogram (RUNW)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only.	Multi-looked, ellipsoid and topography flattened unwrapped interferogram in Range Doppler coordinates.	On pre-defined track/frame

L2 Product	Scope	Description	Granule Size
Geocoded SLC (GSLC)	Global and all channels.	Single Look Complex SAR image on geocoded map coordinate system.	On pre-defined track/frame
Geocoded Nearest-Time Pixel Offsets (GOFF)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only.	Unfiltered and unculled layers of pixel offsets in with different resolutions obtained from incoherent cross correlation and geocoded on map coordinate system.	On pre-defined track/frame
Geocoded Nearest-Time Unwrapped Interferogram (GUNW)	Global. Nearest pair in time and co-pol channels only.	Geocoded, Multi-looked, ellipsoid and topography flattened unwrapped interferogram.	On pre-defined track/frame

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L2 Product	Scope	Description	Granule Size
Geocoded Polarimetric Covariance Matrix (GCOV)	Global and all channels. Single/Dual/Quad pol.	Geocoded, multi-looked polarimetric covariance matrix.	On pre-defined track/frame

Table 2-2 NISAR Data Level Descriptions defined by Science.

Data Level	Description
Level 0A	Unprocessed instrument data with all communications artifacts removed, but without reconstruction of missing data and sorting of samples from the instrument. May still contain bit errors and missing data that needs reconstruction.
Level 0B	Reconstructed, time ordered unprocessed instrument data at original resolution, time ordered.
Level 1	Processed instrument data, focused to full resolution complex images or derived radar parameters including interferometric phase and pixel offsets, in native radar coordinates system.
Level 2	Focused radar imagery or derived radar parameters projected to a map coordinate system.
Level 3	Derived geophysical parameters on geocoded grids with the same or coarser posting as Level 1 or Level 2 products.

2.2 GSLC Overview

The GSLC product is a Level 2 product derived from the Level-1 RSLC product by geocoding the input RSLC into a geocoded map coordinate system such as UTM/ Polar stereographic projection system (Appendix B: Geocoded Product Grids). The geocoding is performed by inverse mapping of the map coordinates with their topographic heights into the radar coordinate system and interpolating the radar signal at the radar location corresponding to the map coordinate. Phase preserving complex interpolation is used to project the data onto a uniformly spaced, north-south/east-west aligned geographic grid. The phase of the GSLC product is flattened with respect to the orbit used in the RSLC processing. The phase flattening removes the topographic phase contribution in the GSLC. Consequently, cross multiplying two GSLC products will result in a interferometric phase flattened interferogram. For more details about the geocoding algorithm please see the NISAR NASA SDS Algorithm Theoretical Basis Document (ATBD) [RD1].

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The spacing of the GSLC product in East and North directions is comparable to the full resolution original RSLC product. Table 2-3 lists the RSLC sampling in slant range and azimuth directions, the ground range sampling in mid-swath and the GSLC sampling in east and north directions for different NISAR acquisition modes. The slant range spacing is obtained assuming 1.2x oversampling of the range bandwidth for the 5, 20 and 40 MHz data. For the 77 MHz data the data are sampled at 1.2x of 80 MHz which is 96 MHz. The ground ranges in Table 2-3 are derived assuming incidence angle of 43.5 degrees in mid swath for 5, 20 and 40 MHz data and assuming 41.7 degrees for half-swath 77 MHz data.

Table 2-3 Posting of GSLC product based on imaging bandwidth.

Range bandwidth (MHz)	RSLC Azimuth sampling (m)	RSLC Range sampling (m)	Ground Range Sampling in mid-Swath (m)	GSLC Posting in Northing (m)	GSLC Posting in Easting (m)
5	~5	~25	~36.3	5	40
20	~5	~6.25	~9.1	5	10
40	~5	~3.12	~4.5	5	5
77	~5	~1.56	~2.27	5	2.5

The GSLC product contains individual binary raster layers representing complex signal return for each polarization layer.

The structure of the GSLC product is described in Section 4. The details of the data elements are given in Section 5. Metadata cubes are discussed in Section 6.

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3 PRODUCT ORGANIZATION

3.1 File Format

All NISAR standard products are in the HDF5 [RD5]. HDF5 is a general-purpose file format and programming library for storing scientific data. The National Center for Supercomputing Applications (NCSA) at the University of Illinois developed HDF to help scientists share data more easily. Use of the HDF library enables users to read HDF files regardless of the underlying computing environments. HDF files are equally accessible in Fortran, C/C++, and other high-level computation packages such as IDL, MATLAB or Python.

The HDF Group, a spin-off organization of the NCSA, is responsible for development and maintenance of HDF. Users should reference The HDF Group website at https://portal.hdfgroup.org/documentation [RD5] to download HDF software and documentation.

HDF5 represents a significant departure from the conventions of previous versions of HDF. The changes that appear in HDF5 provide flexibility to overcome many of the limitations of previous releases. The basic building blocks have been largely redefined and are more powerful but less numerous. The key concepts of the HDF5 Abstract Data Model are Files, Groups, Datasets, Datatypes, Attributes, and Property Lists. The following sections provide a brief description of each of these key HDF5 concepts.

3.1.1 HDF5 File

A File is the abstract representation of a physical data file. Files are containers for HDF5 Objects. These Objects include Groups, Datasets, and named Datatypes.

3.1.2 HDF5 Group

Groups provide a means to organize the HDF5 Objects in HDF5 Files. Groups are containers for other Objects, including other Groups. In that sense, Groups are analogous to directories that are used to categorize and classify files in standard operating systems.

Groups and their nested objects can be accessed using a path-like notation, akin to the notation employed for accessing Unix directories. The root Group is "/". A Group contained in root might be called "/myGroup".

3.1.3 HDF5 Dataset

The Dataset is the HDF5 component that stores user data. Each Dataset associates with a Dataspace that describes the data dimensions, as well as a Datatype that describes the basic unit of storage element. A Dataset can also have Attributes.

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3.1.4 HDF5 Datatype

A Datatype describes a unit of data storage for Datasets and Attributes. Datatypes are subdivided into Atomic and Composite Types.

Atomic Datatypes are analogous to simple basic types in most programming languages. HDF5 Atomic Datatypes include Time, Bitfield, String, Reference, Opaque, Integer, and Float. Each atomic type has a specific set of properties. Examples of the properties associated with Atomic Datatypes are:

- Integers are assigned size, precision, offset, pad byte order, and are designated as signed or unsigned.
- Strings fixed-length byte sequences
- References are constructs within HDF5 Files that point to other HDF5 Objects in the same file.

HDF5 provides a large set of predefined Atomic Datatypes. Table 3-1 lists the Atomic Datatypes that are used in NISAR data products.

Table 3-1. HDF5 Atomic Datatypes

HDF5 Atomic Datatypes	Description
H5T_STD_U8LE	unsigned, 8-bit, little-endian integer
H5T_STD_U16LE	unsigned, 16-bit, little-endian integer
H5T_STD_U32LE	unsigned, 32-bit, little-endian integer
H5T_STD_U64LE	unsigned, 64-bit, little-endian integer
H5T_STD_I8LE	signed, 8-bit, little-endian integer
H5T_STD_I16LE	signed, 16-bit, little-endian integer
H5T_STD_I32LE	signed, 32-bit, little-endian integer
H5T_STD_I64LE	Signed, 64-bit, little-endian integer
H5T_IEEE_F32LE	32-bit, little-endian, IEEE floating point
H5T_IEEE_F64LE	64-bit, little-endian, IEEE floating point
H5T_C_S1	character string made up of one or more bytes

Derived Datatypes are user-defined variants of predefined Atomic Datatypes where the data organization has been modified at the bit-level. Derived data types are particularly useful for representing custom N-bit integers and floating-point numbers.

Composite Datatypes incorporate sets of Atomic Datatypes. Composite Datatypes include Array, Enumeration, Variable Length and Compound.

- The Array Datatype defines a multi-dimensional array that can be accessed atomically.
- Variable Length presents a 1-D array element of variable length. Variable Length Datatypes are useful as building blocks of ragged arrays.
- Compound Datatypes are composed of named fields, each of which may be dissimilar Datatypes. Compound Datatypes are conceptually equivalent to structures in the C programming language.

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Named Datatypes are explicitly stored as Objects within an HDF5 File. Named Datatypes provide a means to share Datatypes among Objects. Datatypes that are not explicitly stored as Named Datatypes are stored implicitly. They are stored separately for each Dataset or Attribute they describe.

The Derived and Compound Datatypes used in NISAR products are reported in **Error! Reference source not found.**.

Table 3-2 NISAR HDF5 Derived and Compound Datatypes

Description	Comments
16-bit little-endian floating point	"binary16" half precision type in IEEE 754-2008 standard. Matches numpy.float16 type in Python. We will refer to this type as H5T_IEEE_F16LE or Float16 in our documents.

H5T_COMPOUND
{
16-bit little-endian floating-point "r";
16-bit little-endian floating-point "i";
}
	Complex numbers made up of two half precision floating point numbers.

H5T_COMPOUND
{
32-bit little-endian floating-point "r";
32-bit little-endian floating-point "i";
}
	Complex numbers made of two single precision floating point numbers.

H5T_COMPOUND
{
64-bit little-endian floating-point "r";
64-bit little-endian floating-point "i";
}
	Complex numbers made of two double precision floating point numbers.

3.1.5 HDF5 Attribute

An Attribute is a small aggregate of data that describes Groups or Datasets. Like Datasets, Attributes are also associated with a particular Dataspace and Datatype. Attributes cannot be subsetted or extended. Attributes themselves cannot have Attributes.

3.2 NISAR File Organization

3.2.1 Groups

All NISAR HDF5 files are organized as groups with no actual data at the root("/") level. Table 3-3 shows the general layout of the HDF5 files that are generated by the NISAR Science Data System. All data are organized under "/science" with data from the L-SAR and S-SAR instruments separated into their own groups.

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Table 3-3 Group organization at the top level of a NISAR HDF5 File

Group Name	Description
/science/LSAR	All science data from the L-SAR instrument is organized under this group
/science/SSAR	All science data from the S-SAR instrument is organized under this group
/science/[L|S]SAR/identification	File level metadata for cataloging, archiving the particular granule

In the nominal baseline, L-SAR and S-SAR data will not appear in the same granule, even if they cover the same geographic area. The rest of the document from this point on describes the layout of the product containing L-SAR data.

3.2.2 File Level Metadata

Global metadata at the file level are currently given as Global Attributes shown in Table 3-4.

Metadata regarding the data in the particular granule are given in the "/science/LSAR/identification" Group. These data are described further in Sec 5.2.

Table 3-4 Global Attributes of GSLC

Attribute	Format	Description	Value
Conventions	string	NetCDF-4 conventions adopted in this product	CF-1.7
title	string	Product title	NISAR L2 GCOV Product
institution	string	Name of producing agency	NASA JPL
mission_name	string	Mission name	NISAR
reference_document	string	Name and version of Product Description Document to use as reference for product	D-102274 NISAR NASA SDS Product Specification Level-2 Geocoded Polarimetric Covariance GCOV
contact	string	Contact information for producer of the product	nisar-sds-ops@jpl.nasa.gov

3.2.3 Variable Metadata (HDF5 Attributes)

NISAR standards incorporate additional metadata that describe each HDF5 Dataset within the HDF5 file. Each of these metadata elements appear in an HDF5 Attribute that is directly associated with the HDF5 Dataset. Wherever possible, these HDF5 Attributes employ names that conform to the Climate and Forecast (CF) conventions.

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Table 3-5 lists the CF names for the HDF5 Attributes that NISAR products typically employ.

Table 3-5. Common variable attributes in HDF5 file.

Attribute	Description
_FillValue	The value used to represent missing or undefined data
description	Miscellaneous information about the data or the methods to generate it
long_name	A descriptive variable name that indicates its content
quality_flag	Names of variable quality flag(s) that are associated with this variable to indicate its quality
units	Unit of data
valid_max	Maximum theoretical value of the variable
valid_min	Minimum theoretical value of the variable

Some HDF5 datasets are populated with statistical attributes. Table 3-6 and Table 3-7 describe statistical attributes added to real- and complex-valued HDF5 datasets, respectively.

Table 3-6. Statistical attributes for real-valued HDF5 datasets.

Attribute	Description
min_value	Minimum value of a real-valued HDF5 dataset
mean_value	Mean value of a real-valued HDF5 dataset
max_value	Maximum value of a real-valued HDF5 dataset
sample_stddev	Sample standard deviation of a real-valued HDF5 dataset

Table 3-7. Statistical attributes for complex-valued HDF5 datasets.

Attribute	Description
min_real_value	Minimum value of the real part of a complex-valued HDF5 dataset
mean_real_value	Mean value of the real part of a complex-valued HDF5 dataset
max_real_value	Maximum value of the real part of a complex-valued HDF5 dataset
sample_stddev_real	Sample standard deviation of the real part of a complex-valued HDF5 dataset
min_imag_value	Minimum value of the imaginary part of a complex-valued HDF5 dataset
mean_imag_value	Mean value of the imaginary part of a complex-valued HDF5 dataset
max_imag_value	Maximum value of the imaginary part of a complex-valued HDF5 dataset
sample_stddev_imag	Sample standard deviation of the imaginary part of a complex-valued HDF5 dataset

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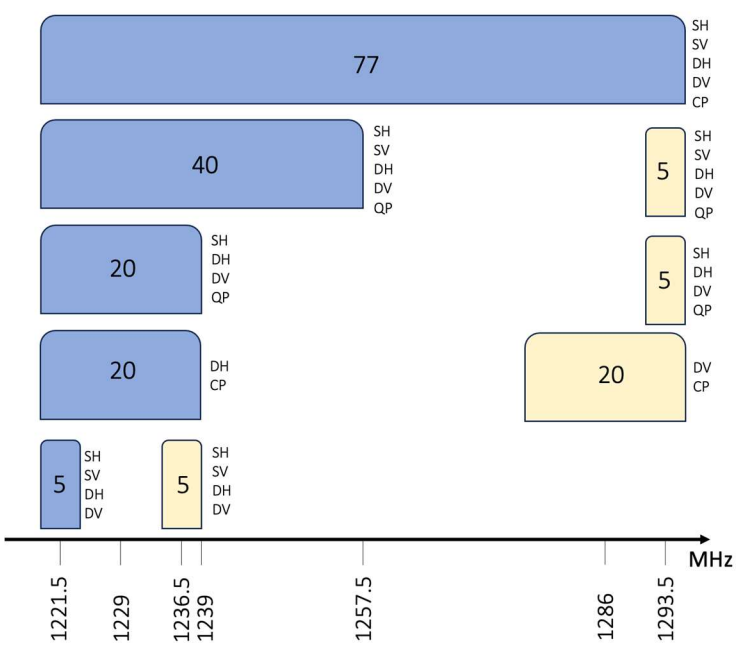
Table 3-8. GSLC HDF5 datasets populated with statistical attributes.

HDF5 Group	HDF5 Datasets	Dataset type
/science/{L/S}SAR/GSLC/grids/frequency{A/B}	HH, HV, VH, VV, RH, RV	Complex-valued

The list of complex-valued HDF5 datasets for the standard GSLC product is given in Table 3-8. Users should note that the existence of the different frequencies (frequency A or B) and polarizations listed in table 3-8 (HH, HV, VH, VV, RH and RV) in a NISAR GSLC product depends on the LSAR instrument's acquisition mode used over specific region. Figure 3-1 visualizes different possibilities of NISAR acquisitions in terms of range bandwidth and polarization of the transmitted and received signal. The LSAR instrument is capable of acquiring SAR data at the total 77 MHz range-bandwidth or at parts of the spectrum as depicted in Figure 3-1. Acquisitions from the full 77 MHz spectrum and acquisitions from lower band of the spectrum (blue color in Fig 3-1) are stored in the "frequencyA" group of the NISAR L-band product (/science/LSAR/GSLC/grids/frequencyA). The SAR data at the higher band of the range spectrum (yellow color in Figure 3-1) are stored in the "frequencyB" group of the L-band product (/science/LSAR/GSLC/grids/frequencyB). Each of the acquisitions in frequencyA or frequencyB groups may be provided in one of the polarization combinations as shown on the right side of each of the frequency band boxes in Figure 3-1. For example a 77 MHz GSLC is expected to only have frequencyA group in the HDF5 file and depending on the mode of acquisition may contain one of the polarization combinations. Many 77 MHz data are acquired over cryosphere targets with HH polarization. A GSLC product at 20+5 MHz acquisition in dual polarization H-transmit and V-receive, contains frequencyA and frequencyB groups with HH and HV complex datasets for each of those groups. Similarly, one can extract other frequency or polarization combinations from Figure 3-1.



Frequency A
Frequency B

SH: Single-Polarization H-transmit and H-receive (HH)
SV: Single-Polarization V-transmit and V-receive (VV)
DH: Dual-Polarization H-transmit and H- and V-receive (HH and HV)
DV: Dual-Polarization V-transmit and V- and H-receive (VV and VH)
QP: Quad-Polarization H- and V-transmit and H- and V-receive (HH,HV,VH,VV)
CP: Compact Polarization (RH, RV)

L-band mode Bandwidth [MHz]	Center frequency [MHz]	
	Frequency A	Frequency B
77	1257.5	NA
40+5	1239	1293.5
20+5	1229	1293.5
20+20	1229	1286
5+5	1221.5	1236.5

Figure 3-1 A visualization of possible NISAR L-band science acquisition modes.

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3.2.4 Georeferenced HDF5 Datasets

NISAR L2 products contain georeferenced Datasets where the georeferencing information is provided in accordance with Climate and Forecast 1.7 (CF 1.7) conventions.

CF conventions require a "grid mapping" dataset which describes the coordinate system associated with the georeferenced Dataset. For NISAR L2 products, this grid mapping is represented by the Dataset "projection", which is included under the same Group as each georeferenced Dataset. Accordingly, each georeferenced Dataset contains an Attribute “grid_mapping”, whose value is always hard-coded to the string "projection" (see Table 3-9).

The value of the "projection" Dataset is set to the European Petroleum Survey Group (EPSG) code of the associated georeferenced Dataset. The "projection" Dataset has Attributes with additional grid mapping information (see Table 3-10). More information about the projections used to represent NISAR L2 products is provided in Appendix B: Geocoded Product Grids.

In addition to a grid mapping dataset, CF conventions use HDF5 Dimension Scales [RD6] to associate coordinates to each georeferenced Dataset. The Dimension Scales employed in NISAR L2 products are the "xCoordinates", "yCoordinates", and "heightAboveEllipsoid" Datasets, which represent the horizontal X- and Y-coordinates, and elevation Z-coordinates, respectively, and are located within the same Group as the associated georeferenced Dataset. These are one dimensional (1-D) vectors with lengths matching the associated Dataset's dimensions; each vector element corresponds to the grid-mapping location at the center of the georeferenced array pixels. "heightAboveEllipsoid" is only included for three-dimensional (3-D) georeferenced Datasets.

The complete listing of all georeferenced HDF5 Datasets within the GSLC product is given in Table 3-11. Note that the 3-D georeferenced Datasets are contained in the "radarGrid" Group; they are *metadata cubes* which represent the radar geometry in a compact form. Sec. 4.4.5 contains information about the “radarGrid” Group; Sec. 6 describes metadata cubes and their usage.

Table 3-9. Geolocation attributes for georeferenced HDF5 Datasets.

Attribute	Description	Value
grid_mapping	Grid mapping Dataset name	projection

Table 3-10 Attributes of the HDF5 Dataset "projection" containing the grid mapping.

Attribute	Description
ellipsoid	Projection ellipsoid
epsg_code	Projection EPSG code
false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection

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false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection
grid_mapping_name	Grid mapping variable name
inverse_flattening	Inverse flattening of the ellipsoidal figure
latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection
longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum
semi_major_axis	Semi-major axis
spatial_ref	Spatial reference
utm_zone_number	UTM zone number

Table 3-11 List of GSLC georeferenced Datasets.

HDF5 Group	HDF5 Datasets	Array Dimension
/science/LSAR/GSLC/grids/frequency[A/B]	HH/HV/VV/VH	2-D
/science/LSAR/GSLC/metadata/radarGrid	slantRange zeroDopplerAzimuthTime incidenceAngle, losUnitVectorX losUnitVectorY alongTrackUnitVectorX alongTrackUnitVectorY elevationAngle groundTrackVelocity	3-D

3.3 Granule Definition

NISAR GSLC Granules will conform to the Tiling Scheme being developed for the mission and are expected to have a ground footprint of approximately 240 km x 240 km. The exact size may differ depending on the frame location. Each GSLC product is provided on a predefined fixed geocoded output grid.

3.4 File Naming Convention

NISAR GSLC Granule names will conform to the SDS L-SAR Product File Naming Conventions [RD3]. Accordingly, the naming convention for GSLC has the following syntax:

Syntax:

NISAR_*IL_PT_PROD_CYL_REL_P_FRM_MODE_POLE_S_StartDateTime_EndDateTime_CRID_A_C_LOC_CTR*.EXT

Where:

- NISAR – 5 char for mission: **NISAR**
- *I* – 1 char for Instrument: **L** for L-SAR, **S** for S-SAR
- *L* – 1 char for Level: **1** or **2** or **3**
- *PT* – 2 char for Processing Type:

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    - **PR –** Production
    - **UR –** Urgent Response
    - **OD – Science** On-Demand
- *PROD* – 4 chars for Product Identifier: **GSLC**
- *CYL* – 3 chars for CycLe number in the mission, each cycle represents 12 days, zero padded, starting at 001. **Error! Reference source not found.**
- *REL* – 3 chars for RELative orbit track number within a cycle, resets to 1 with a cycle number increment, zero padded.Valid values: 001-173. **Error! Reference source not found.**
- *P* – 1 char for direction of movement of the satellite at the time of imaging **Error! Reference source not found.**:
    - **A** for Ascending
    - **D** for Descending
- *FRM* – 3 chars for track frame number, a segment of an orbital track corresponding to the product, zero padded Valid Values: 001-176 on each track. See sec **Error! Reference source not found.**
- *MODE* – 4 chars for Bandwidth Mode Code of Primary and Secondary Bands:
    - **40, 20, 77, 05, or 00** (only if the secondary band is missing)
- *POLE* – 4 chars for Polarization of the data for the primary and secondary bands. Each band uses a two character code among the following:
    - **SH** = HH – Single Polarity (H transmit and receive)
    - **SV** = VV – Single Polarity (V transmit and receive)
    - **DH** = HH/HV – Dual Polarity (H transmit)
    - **DV** = VV/VH – Dual Polarity (V transmit)
    - **CL**= LH/LV – Compact Polarity (Left transmit)
    - **CR** = RH/RV – Compact Polarity (Right transmit)
    - **QP** = HH/HV/VV/VH – Quad Polarity
    - **NA** if band does not exist

  For example, a "quasi-quad" polarization mode would be noted **DHDV** while a "quasi-dual" polarization mode would be noted **SHSV**.
- **S – 1 char for source of data for the product**
    - **A = Acquired source of the observation, single mode**
    - **M = Mixed source of observations, mixed mode**
- *StartDateTime* – 15 chars for Radar Start Time of the data processed as zero Doppler contained in the file as YYYYMMDDTHHMMSS, UTC
- *EndDateTime* – 15 chars for Radar End Time of the data processed as zero Doppler contained in the file as YYYYMMDDTHHMMSS, UTC
- *CRID* – 6 chars for Composite Release Identifier
    - Format of EPMMmm. See section **Error! Reference source not found.**
- *A* – 1 char for Product Accuracy or Fidelity of the Orbit Ephermis and Radar Pointing:
    - **P**, **M**, **N**, or **F**, See section **Error! Reference source not found.**
- *C* – 1 char as Coverage Indicator: **F** for Full or **P** for Partial. See section **Error! Reference source not found.**
- *LOC* – 1 char to represent the location of the Science Data System. J for JPL, recommends N for NRSC. See section 4.6
- *CTR* – 3 chars for Product Counter (zero padded). See section **Error! Reference source not found.**
- *EXT* – 1 to n chars for Extension: **h5, met, log**

NOTES:
- 17 spacing characters: 16 for underscores plus one period
- Total number of characters: **91** excluding extension

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Examples:
NISAR_L2_PR_GSLC_001_005_A_219_2005_DHDH_A_20220104T182346_20220104T183426_P01101_M_F_J_001.h5
NISAR_L2_PR_GSLC_001_005_A_219_4005_DHDV_A_20220104T182346_20220104T183426_P01101_M_F_J_001.h5NISAR_L2_PR_GSLC_001_005_A_219_8000_SHNA_A_20220104T182346_20220104T183426_P01101_M_F_J_001.h5

3.5 Temporal Organization

Temporal organization is not specifically applicable to the GSLC product, although it is generally arranged in order of increasing azimuth time.

3.6 Spatial Organization

The L2 data are arranged on a uniformly spaced, North-up and West-left grid – i.e., decreasing North or Y coordinate in the row direction and increasing East or X coordinate in the column direction following the row-major order convention of representing 2D raster arrays.

3.7 Spatial Sampling and Resolution

Some salient features of the output grid for the GSLC product are:

1. The top-left corner of the top-left pixel will correspond to the same geographic coordinate for all imagery layers in an L-SAR GSLC product – frequency A and frequency B.
2. The main (frequency A) and auxiliary (frequency B) bands of L-SAR data will share an exact integer scaling relationship to allow for easy inter-comparison (Table 2-3).
3. Pixel-is-area convention is used to tag the raster layers with coordinate information and the origin of the geocoded grid is the top-left corner of the top-left pixel of the grid. (see Figure 3-2 and Appendix B: Geocoded Product Grids).

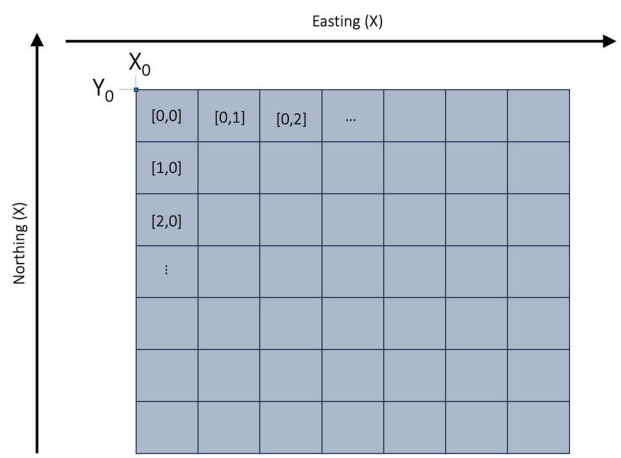

Figure 3-2 Spatial grid convention for the NISAR geocoded products.

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3.7.1 Mosaicking

The spatial sampling of the output grid has been designed to facilitate along-track mosaicking of contiguous GSLC product granules. See Appendix B: Geocoded Product Grids for details on common output grid used for all L2 products.

Note that GSLC products generated from L1 RSLC products with different center frequencies (Figure 3-1) cannot be mosaicked for applications that expect phase continuity.

3.7.2 Partially Compressed Data

Partially compressed (processed) data in RSLC products will be provided in the GSLC products. In case of constant PRF acquisitions, the RSLC products will contain multiple sub-swaths (4 or 5 sub-swaths) with fully focused SAR data within each sub-swath and with partially focused SAR data between the different sub-swaths. The SAR complex imagery in the GSLC product will contain fully and partially focused data. However, a mask of the sub-swaths geocoded to the same grid of the geocoded SAR data is provided to the users so that they can distinguish between the fully focused data from the partially focused data. See section 4.4 for more information about the subswath mask.

3.8 Cloud Optimized HDF5

NISAR science data products utilize several special features of the HDF5 format to optimize file sizes and enable high-performance read access in a cloud environment. A key challenge of cloud data access is the latency associated with calls to the cloud storage API, so the following strategies are used to minimize the number of cloud API calls needed per byte of data read:

- Chunks: Large datasets within the products use chunked storage. Every read operation thus fetches at least one entire chunk of data. The chunk size is nominally 512x512 pixels, though the precise chunk dimensions should be obtained using the H5Pget_chunk method of the HDF5 C API (or its equivalent in other language bindings).
- Compression: Data are written using a compression filter, minimizing the amount of data stored and hence transferred over the network. The HDF5 API handles decompression automatically.
- Paging: Files are created with the “paged” file space strategy (H5F_FSPACE_STRATEGY_PAGE in the HDF5 C API). These pages serve as the basic unit of allocation within the file. The page size is chosen larger than the chunk size so that both a chunk of data and its HDF5-internal metadata can be read in a single cloud API call. This parameter may be queried using the H5Pget_file_space_page_size method of the HDF5 C API.

Software that reads NISAR products stored on the cloud should take heed of the following recommendations:

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- Set the page buffer size to a multiple of the file space page size using H5Pset_page_buffer_size in the HDF5 C API. This enables caching logic that reduces the number of cloud API calls in the file driver.
- Implement chunk-aligned data access patterns. Reads in multiples of the chunk size (and aligned with chunk boundaries) are most efficient.
- If other access patterns are desired, try setting the read cache large enough to hold all the chunks that may be re-read. For example, line-by-line access can still be efficient if the read cache is large enough to hold N lines, where N is the chunk dimension. That way lines can be read from the cache instead of fetching the same set of chunks N times over the network. The cache size may be set globally using the H5Pset_cache or locally with the H5Pset_chunk_cache methods of the HDF5 C API.

Note that, in general, these optimizations require knowledge of the file contents. Therefore, the most robust approach is to open the file, inspect the contents (e.g., chunk size, page size, and dataset dimensions) and then re-open the file with optimal parameters.

4 LEVEL 2 GEOCODED SINGLE LOOK COMPLEX PRODUCT

In this section, we briefly describe the layout of GSLC data and associated metadata in the NISAR HDF5 file. The GSLC product contains imagery layers as Digital Numbers (DNs) with secondary layer LUTs to convert to beta0, sigma0 and gamma0 with respect to the WGS84 Ellipsoid. In this section, we focus on the organization of L-SAR instrument data under the Group name "/science/LSAR".

4.1 Dimensions and Shapes of Data

Information on the dimensions and shapes of the data items in various data tables is described as part of the metadata (Sec 5.1). This information is useful both as part of the product identification and for setting up further processing, i.e., dimensioning arrays.

4.2 Product Identification

Information needed to identify this particular product is given under the Group "/science/LSAR/identification" (Sec 5.2). This includes information such as orbit number, track-frame number, acquisition times, a polygon representing the bounding box of the included imagery in geographic coordinates, and product version.

4.2.1 Composite Release Identifier

The Composite Release Identifier (CRID) is a global version identifier documenting the algorithms and the overall status of the science data system used to generate the product. The CRID follows the format *EPMMmp* where:

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- **E (Environment)**: a single character representing the environment or the venue where the product was generated. It can assume the values:
  - *A*: if the product was generated in the Algorithm Development environment
  - *D*: if the product was generated in the Development environment
  - *P*: if the product was generated in the Production environment
  - *T*: if the product was generated in the Integration and Test (I&T) environment
- **P (Mission Phase)**: a single numerical digit indicating the mission phase in which the product was generated. It can assume the following values:
  - *0*: for pre-launch (Phase D)
  - *1*: for primary science phase operations (Phase E)
  - *2*: extended mission (Phase E)
  - *3*: post-operations (Phase F), decommissioning, end of mission processing
- **MM (Major Release)**: two numeric digits monotonically increasing between 0 and 99. The Major Release resets to zero upon a change in the Mission Phase identifier. A change in the Major Release indicates a major change in the products i.e., a change to one or more algorithms or to the processing rules having a significant impact on the science content of the product. The Major Release stands as a composite of the versions of all the algorithms used in the science data production systems. Individual algorithm versions are allocated in the product metadata.
- **m (Minor Release):** a single numeric digit increasing monotonically between 0 and 1 indicating a minor update to the product and/or the data system. A change in the Minor Release identifier indicates minor algorithm changes (e.g., bug fixes, small functional updates) that do not have a significant impact on the product. The Minor Release identifier resets to zero upon every update to the Major Release identifier
- **p (Patch Release)**: a single numerical digit monotonically increasing between 0 and 1. A change in the Patch Release identifier indicates an update to the science data system software that has undergone the System Deployment Review to fix a critical bug. The Patch Release resets to zero upon updates to the Major Release or Minor Release identifiers.

4.3 Radar Imagery

The primary data elements for the granule are in the group "/science/LSAR/GSLC/grids" with subgroups for frequencyA and frequencyB (if present). Imagery layers are further organized as individual 2D datasets by polarization (TxRx) under ../frequency[A|B]. The details of the data elements are given in Section 5.3.

4.4 Mask layer

The mask layer is provided as an unsigned byte (8 bits) HDF5 Dataset located at "/science/LSAR/GSLC/grids/frequency[A|B]/mask. It has the same geographic grid as the GSLC imagery. Each pixel of the mask layer is associated with the GSLC image indexes. The mask layer in GSLC is a geocoded raster based on the interpolation window with respect to the valid sample data in RSLC products which are described by validSamplesSubSwath[i], i in [1,5], in RSLC specifications. Possible mask values for mask layer include:

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- “0”: Invalid or partially focused – the interpolation window contains at least one pixel in the “invalid” or “partially focused” region of the RSLC.
- “1” to “5”: Valid – all the pixels in the interpolation window falls in the fully focused part of the RSLC image. The mask value indicates the sub-swath number from which the radar samples came from, i.e., sub-swath 1 to 5. NISAR RSLC products formed from SweepSAR acquisitions with constant PRF will contain 4 to 5 sub-swaths. The RSLC products focused from Dithered Pulse Repetition Frequency (PRF) acquisitions, will contain only one sub-swath to cover the scene and accordingly the value of the valid portion of the mask layer in the GSLC metadata will be 1.
- “255”: Fill Value - The geographical extents of the GSLC pixel are outside of the RSLC image bounds.

4.5 Radar Metadata

Radar metadata needed to interpret the amplitude and phase information, as well as the geolocation of the imagery are organized under the folder “/science/LSAR/GSLC/metadata”.

4.5.1 Timing convention

The NISAR GSLC product uses Coordinated Universal Time (UTC) as its timing convention. All time-dependent radar metadata within the GSLC product, including metadata cubes in the radar grid and timing correction LUTs, are tagged with UTC.

4.5.2 Calibration

The subgroup “calibrationInformation” contains two major types of information: radiometric calibration and radar information. The complete list of calibration information fields is given in Section 5.4.

4.5.2.1 Calibration Information

Lookup tables (LUT), common to all frequencies and polarizations as these are purely a function of imaging geometry, are organized under the subgroup “calibrationInformation/geometry”. The radar imagery themselves are provided as Digital Numbers (DNs). LUTs are provided to transform the DNs to beta0, sigma0 and gamma0 (with respect to the reference ellipsoid) as follows:

```
beta0 = abs(RSLC)^2 / beta0_LUT^2
sigma0 = abs(RSLC)^2 / sigma0_LUT^2
gamma0 = abs(RSLC)^2 / gamma0_LUT^2
```

These LUTs are provided as a sparse grid in map coordinates. Values at any geographical location can be obtained using simple 2D interpolation (bilinear or higher order).

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4.5.2.2 Radar Information

Complex two-way antenna patterns and noise-equivalent backscatter (neb) are organized by frequency and polarization. These datasets are provided on a sparse grid in map coordinates and values of interest at any geographical location can be estimated using simple 2D interpolation (bilinear or higher order).

4.5.3 Processing Information

The metadata related to processing parameters, algorithms, and inputs used to produce the product are given in Section 5.5.

4.5.3.1 Parameters

Processing parameters such as Doppler centroid are organized by frequency under the subgroup "processingInformation/parameters". Common parameters such as reference terrain height and chirp weighting parameters are also included in this subgroup. All processing parameters that vary spatially are organized on low resolution geocoded grids to allow for easy lookup based on map coordinates. This subgroup also includes flags identifying different possible corrections applied to improve the geolocation accuracy of the product. In the current version of the product, the geolocation is corrected for ionospheric range delay and dry tropospheric range delay. The ionospheric delay is estimated using GNSS-based TEC data and corrected during the geocoding process. The dry tropospheric delay is computed using a static model [RD1] and corrected during focusing the RSLC product. The subgroup also includes a flag for possible radio frequency interference (RFI) correction ("rfiCorrectionApplied") applied to the input RSLC product.

4.5.3.2 Algorithm Information

The processing algorithm information is provided in the subgroup "processingInformation/algorithms/". It includes the software version ("softwareVersion"), which is the version of the ISCE3 software that was used to generate the product, and the list of the algorithms employed in the product processing.

4.5.3.3 Inputs

The key input file – L1 RSLC granule, orbit, DEM source description, and configuration files are tracked and listed under the subgroup "processingInformation/inputs".

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4.5.4 Other Radar Metadata

Section 5.6 includes the orbit ephemeris used for generating the GSLC under a subgroup named "metadata/orbit" and the attitude under a subgroup named "metadata/attitude".

4.5.4.1 Orbit

The orbit ephemeris used for generating the GSLC product can be found under a subgroup named "orbit". This group includes time-tagged antenna phase center position and velocity vectors in Earth Centered Earth Fixed (ECEF) cartesian coordinates. In nominal operations, this would be the MOE state vectors that were used by the L2 processor.

4.5.4.2 Attitude

The attitude state vectors used for generating the GSLC product can be found under a subgroup named "attitude". This group includes time-tagged quaternions and Euler Angles representing the slant range plane from the antenna phase center in Earth Centered Earth Fixed (ECEF) cartesian system. In nominal operations, this would be the restituted attitude state vectors that were used by the L2 processor.

4.5.5 Radar Grid

Section 5.7 contains information describing the radar geometry of the sensor during data taking in the group "/science/LSAR/GSLC/metadata/radarGrid/". This information is given in the form of data cubes, referred to as *radar grid cubes*, that are organized over a three-dimensional geographic grid. The representation as data cubes, rather than two-dimensional rasters, is used to reduce the amount of space required to store radar geometry values within NISAR L2 products. This is possible because each radar grid cube contains slowly-varying values in space that can be described by a low-resolution three-dimensional grid with sufficient accuracy.

These values, however, are usually required at the terrain height, often characterized by a fast-varying surface representing the local topography. A higher-resolution DEM can then be used to interpolate radar grid cubes and generate high-resolution maps of the corresponding radar geometry variable.

Radar grid cubes (for geocoded products) are provided in the same coordinate system as the product imagery with similar extents (bounding box) but coarser pixel spacing. The three-dimensional geographic grid is defined by the HDF5 datasets "xCoordinates" (defining the east component), "yCoordinates" (north component), and "heightAboveEllipsoid" (height above the WGS84 ellipsoid), common to all radar grid cubes, and following CF conventions 1.7.

Radar grid cubes provide the following list of radar geometry information in the associated HDF5 datasets:

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1. The zero-Doppler radar grid is defined through the datasets “slantRange” and “zeroDopplerAzimuthTime”, which contain respectively the range position in meters and the zero-Dopper azimuth time in seconds for each point of the geographic grid.
2. The line-of-sight (LOS) unit vector, i.e., the vector from the target to the sensor, is defined by the datasets “losUnitVectorX” and “losUnitVectorY” which contain respectively the east and north components of the LOS unit vector in the east-north-up (ENU) coordinate system for each point of the geographic grid. Note that the third (“up”) component of the LOS unit vector e_z is not provided in the product as it can be simply derived from the other two components as
$$e_z = \sqrt{1 - e_x^2 - e_y^2}$$
3. The along-track unit vector represents the projection of the along-track vector at the ground height. It is defined by the datasets “alongTrackUnitVectorX” and “alongTrackUnitVectorY” containing respectively the east and north components of the along-track unit vector in UTM coordinates.
4. The incidence angle, i.e., the angle between the LOS vector and the normal to the ellipsoid at the target height, is given by the dataset “incidenceAngle.
5. The elevation angle, defined as the angle between the LOS vector and the normal to the ellipsoid at the sensor, is provided as “elevationAngle”.
6. The ground track velocity which contains the absolute value of the platform velocity scaled at the target height is given as “groundTrackVelocity”.

5 PRODUCT SPECIFICATION

The following subsections describe the structure of the HDF5 product, the data and metadata contained in the product, their data types, attributes and units when possible. In all the following tables the data type “string” is used for “fixed-length byte arrays” which are used to store string data types.

5.1 Dimensions and Shapes

To simplify the description of the layout of data within the HDF5 file, we will use a table of dimensions and shapes to represent the relationship between similarly sized datasets. The entries in this table do not present actual datasets in the HDF5. This table is meant to be a guide to interpreting the shapes of the datasets in subsequent subsections.

Table 5-1 Table of dimensions and shapes in GSLC product

Name	Shape	Description
scalar	scalar	scalar values
numberOfDatatakes	scalar	number of datatakes in product
numberOfObservations	scalar	number of observations in product

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numberOfFrequencies	scalar	Number of L-SAR frequencies in product
numberOfFrequencyAPolarizations	scalar	Number of polarization layers associated with L-SAR frequency A
frequencyAWidth	scalar	Number of pixels in all L-SAR frequency A imagery datasets
numberOfSourceDataFrequencyAPolarizations	scalar	Number of polarization layers associated with L-SAR frequency A
frequencyALength	scalar	Number of lines in all L-SAR frequency A imagery datasets
complexDataFrequencyAShape	(frequencyALength, frequencyAWidth)	Shape associated with L-SAR frequency A imagery datasets
realDataFrequencyAShape	(frequencyALength, frequencyAWidth)	Shape associated with L-SAR frequency A real-valued raster layers
numberOfFrequencyBPolarizations	scalar	Number of polarization layers associated with L-SAR frequency B
numberOfSourceDataFrequencyBPolarizations	scalar	Number of polarization layers associated with L-SAR frequency B
frequencyBWidth	scalar	Number of pixels in all L-SAR frequency B imagery datasets
frequencyBLength	scalar	Number of lines in all L-SAR frequency B imagery datasets
complexDataFrequencyBShape	(frequencyBLength, frequencyBWidth)	Shape associated with L-SAR frequency B imagery datasets
realDataFrequencyBShape	(frequencyBLength, frequencyBWidth)	Shape associated with L-SAR frequency B real-valued raster layers
radarCubeShape	(radarCubeHeight, radarCubeLength, radarCubeWidth)	Shape associated with metadata cubes
radarGridShape	(radarCubeLength, radarCubeWidth)	Shape associated with metadata 2D layers
radarCubeHeight	scalar	Height dimension of the metadata cube
radarCubeLength	scalar	Length dimension of the metadata cube
radarCubeWidth	scalar	Width dimension of the metadata cube
dopplerCentroidLength	scalar	Length dimension of Doppler centroid grid
dopplerCentroidWidth	scalar	Length dimension of Doppler centroid grid
dopplerCentroidShape	(dopplerCentroidLength, dopplerCentroidWidth)	Shape of the Doppler centroid grid
calibrationLength	scalar	Length of calibration LUTs
calibrationWidth	scalar	Width of calibration LUTs
calibrationScaleShape	(calibrationTimeLength, calibrationSlantRangeWidth)	Shape of calibration LUTs
antennaPatternComplexShape	(calibrationTimeLength, calibrationSlantRangeWidth)	Shape of antenna pattern datasets
crosstalkComplexShape	scalar	Shape of crosstalk datasets
timingCorrectionLength	scalar	Length of timing correction LUTs
timingCorrectionWidth	scalar	Width of timing correction LUTs
timingCorrectionShape	(timingCorrectionLength, timingCorrectionWidth)	Shape of timing correction LUTs

This document has been reviewed and determined not to contain export controlled technical data.

sourceDataDopplerCentroidTimeLength	scalar	Length dimension of Doppler centroid grid corresponding to the source data
sourceDataDopplerCentroidSlantRangeWidth	scalar	Length dimension of Doppler centroid grid corresponding to the source data
sourceDataDopplerCentroidShape	(sourceDataDopplerCentroidTimeLength, sourceDataDopplerCentroidSlantRangeWidth)	Shape of the Doppler centroid grid corresponding to the source data
orbitListLength	scalar	Number of orbit state vectors
orbitShape	(orbitListLength, 3)	Shape of orbit state vector triplets dataset
attitudeListLength	scalar	Number of attitude state vectors
attitudeQuaternionShape	(attitudeListLength, 4)	Shape of attitude quaternion dataset
attitudeShape	(attitudeListLength, 3)	Shape of attitude Euler angle triplets dataset
chirpWeightingFrequencyLength	scalar	Shape associated with 1D filter representations in frequency domain
numberOfInputL1Files	scalar	Number of input L1 SLC granules
numberOfInputOrbitFiles	scalar	Number of input orbit files
numberOfInputTecFiles	scalar	Number of input total electon content (TEC) files
numberOfInputConfigFiles	scalar	Number of input configuration files

This document has been reviewed and determined not to contain export controlled technical data.

5.2 Product Identification

Table 5-2 NISAR HDF5 variables used for product identification

Product Identification Variables	
/science/LSAR/identification/absoluteOrbitNumber	
Type: UInt32	**Shape: scalar**
Description: Absolute orbit number	
/science/LSAR/identification/trackNumber	
Type: UInt32	**Shape: scalar**
Description: Track number	
/science/LSAR/identification/frameNumber	
Type: UInt16	**Shape: scalar**
Description: Frame number	
/science/LSAR/identification/missionId	
Type: string	**Shape: scalar**
Description: Mission identifier	
/science/LSAR/identification/processingCenter	
Type: string	**Shape: scalar**
Description: Data processing center	
/science/LSAR/identification/productType	
Type: string	**Shape: scalar**
Description: Product type	
/science/LSAR/identification/granuleId	
Type: string	**Shape: scalar**
Description: Unique granule identification name	
/science/LSAR/identification/productDoi	
Type: string	**Shape: scalar**
Description: Digital Object Identifier (DOI) for the product	
/science/LSAR/identification/productVersion	
Type: string	**Shape: scalar**
Description: Product version which represents the structure of the product and the science content governed by the algorithm, input data, and processing parameters	
/science/LSAR/identification/productSpecificationVersion	
Type: string	**Shape: scalar**
Description: Product specification version which represents the schema of this product	
/science/LSAR/identification/lookDirection	
Type: string	**Shape: scalar**
Description: Look direction, either "Left" or "Right"	
/science/LSAR/identification/orbitPassDirection	
Type: string	**Shape: scalar**
Description: Orbit direction, either "Ascending" or "Descending"	
/science/LSAR/identification/zeroDopplerStartTime	
Type: string	**Shape: scalar**
Description: Azimuth start time (in UTC) of the product in the format YYYY-mm-ddTHH:MM:SS.sssssssss	
/science/LSAR/identification/zeroDopplerEndTime	
Type: string	**Shape: scalar**
Description: Azimuth stop time (in UTC) of the product in the format YYYY-mm-ddTHH:MM:SS.sssssssss	
/science/LSAR/identification/plannedDatatakeId	

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Type: string		**Shape: (numberOfDatatakes)**
Description: List of planned datatakes included in the product		
/science/LSAR/identification/plannedObservationId		
Type: string		**Shape: (numberOfObservations)**
Description: List of planned observations included in the product		
/science/LSAR/identification/isUrgentObservation		
Type: string		**Shape: scalar**
Description: Flag indicating if observation is nominal ("False") or urgent ("True")		
/science/LSAR/identification/listOfFrequencies		
Type: string		**Shape: (numberOfFrequencies)**
Description: List of frequency layers available in the product		
/science/LSAR/identification/diagnosticModeFlag		
Type: UByte		**Shape: scalar**
Description: Indicates if the radar operation mode is a diagnostic mode (1-2) or DBFed science (0): 0, 1, or 2		
/science/LSAR/identification/productLevel		
Type: string		**Shape: scalar**
Description: Product level. L0A: Unprocessed instrument data; L0B: Reformatted, unprocessed instrument data; L1: Processed instrument data in radar coordinates system; and L2: Processed instrument data in geocoded coordinates system		
/science/LSAR/identification/isGeocoded		
Type: string		**Shape: scalar**
Description: Flag to indicate if the product data is in the radar geometry ("False") or in the map geometry ("True")		
/science/LSAR/identification/boundingPolygon		
Type: string		**Shape: scalar**
Description: OGR compatible WKT representing the bounding polygon of the image. Horizontal coordinates are WGS84 longitude followed by latitude (both in degrees), and the vertical coordinate is the height above the WGS84 ellipsoid in meters. The first point corresponds to the start-time, near-range radar coordinate, and the perimeter is traversed in counterclockwise order on the map. This means the traversal order in radar coordinates differs for left-looking and right-looking sensors. The polygon includes the four corners of the radar grid, with equal numbers of points distributed evenly in radar coordinates along each edge		
	ogr_geometry	polygon
	epsg	4326
/science/LSAR/identification/processingDateTime		
Type: string		**Shape: scalar**
Description: Processing date and time (in UTC) in the format YYYY-mm-ddTHH:MM:SS		
/science/LSAR/identification/radarBand		
Type: string		**Shape: scalar**
Description: Acquired frequency band, either "L" or "S"		
/science/LSAR/identification/platformName		
Type: string		**Shape: scalar**
Description: Name of the platform used to collect the remote sensing data provided in this product		
/science/LSAR/identification/instrumentName		
Type: string		**Shape: scalar**
Description: Name of the instrument used to collect the remote sensing data provided in this product		
/science/LSAR/identification/processingType		
Type: string		**Shape: scalar**
Description: Nominal (or) Urgent (or) Custom (or) Undefined		
/science/LSAR/identification/isDithered		
Type: string		**Shape: scalar**
Description: "True" if the pulse timing was varied (dithered) during acquisition, "False" otherwise.		
/science/LSAR/identification/isMixedMode		
Type: string		**Shape: scalar**
Description: "True" if this product is a composite of data collected in multiple radar modes, "False" otherwise.		
/science/LSAR/identification/isFullFrame		
Type: string		**Shape: scalar**

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Description: "True" if the product fully covers a NISAR frame, "False" if partial coverage		
	frameCoveragePercentage	Percentage of NISAR frame containing processed data
	thresholdPercentage	Threshold percentage used to determine if the product is full frame or partial frame
/science/LSAR/identification/compositeReleaseId		
Type: string		**Shape: scalar**
Description: Unique version identifier of the science data production system		
/science/LSAR/identification/isJointObservation		
Type: string		**Shape: scalar**
Description: "True" if any portion of this product was acquired in a joint observation mode (e.g., L-band and S-band simultaneously), "False" otherwise		

5.3 Radar Imagery

Table 5-3 NISAR HDF5 variables related to SAR imagery

Product Imagery Variables		
/science/LSAR/GSLC/grids/frequencyA/listOfPolarizations		
Type: string		**Shape: (numberOfFrequencyAPolarizations)**
Description: List of processed polarization layers with frequency A		
/science/LSAR/GSLC/grids/frequencyA/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GSLC/grids/frequencyA/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GSLC/grids/frequencyA/rangeBandwidth		
Type: Float64		**Shape: scalar**
Description: Processed range bandwidth in hertz		
	units	hertz
/science/LSAR/GSLC/grids/frequencyA/azimuthBandwidth		
Type: Float64		**Shape: scalar**
Description: Processed azimuth bandwidth in hertz		
	units	hertz
/science/LSAR/GSLC/grids/frequencyA/centerFrequency		
Type: Float64		**Shape: scalar**
Description: Center frequency of the processed image in hertz		
	units	hertz
/science/LSAR/GSLC/grids/frequencyA/slantRangeSpacing		
Type: Float64		**Shape: scalar**
Description: Slant range spacing of grid. Same as difference between consecutive samples in slantRange array		
	units	meters
/science/LSAR/GSLC/grids/frequencyA/zeroDopplerTimeSpacing		
Type: Float64		**Shape: scalar**
Description: Time interval in the along-track direction for raster layers. This is same as the spacing between consecutive entries in the zeroDopplerTime array		
	units	seconds

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/GSLC/grids/frequencyA/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/grids/frequencyA/xCoordinates		
Type: Float64		**Shape: (frequencyAWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/grids/frequencyA/yCoordinates		
Type: Float64		**Shape: (frequencyALength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/grids/frequencyA/mask		
Type: UByte		**Shape: (frequencyALength, frequencyAWidth)**
Description: Mask indicating the subswath number representing valid GSLC samples. Each GSLC pixel is assumed valid if all the pixels in the interpolation window are fully focused in the input RSLC. A value of `0` indicates that at least one RSLC pixel in the interpolation window is partially focused or invalid. Pixels outside of the radar acquisition extent are filled with the value `255`.		
	grid_mapping	projection
	long_name	Valid samples subswath mask
	valid_min	0
	_FillValue	255
/science/LSAR/GSLC/grids/frequencyA/HH		
Type: CFloat32		**Shape: (frequencyALength, frequencyAWidth)**
Description: Focused SLC image (HH)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points

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	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image HH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyA/HV		
Type: CFloat32		**Shape: (frequencyALength, frequencyAWidth)**
Description: Focused SLC image (HV)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image HV
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyA/VH		
Type: CFloat32		**Shape: (frequencyALength, frequencyAWidth)**
Description: Focused SLC image (VH)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image VH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyA/VV		
Type: CFloat32		**Shape: (frequencyALength, frequencyAWidth)**
Description: Focused SLC image (VV)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image VV
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyA/RH		
Type: CFloat32		**Shape: (frequencyALength, frequencyAWidth)**
Description: Focused SLC image (RH)		

This document has been reviewed and determined not to contain export controlled technical data.

	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image RH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyA/RV		
Type: CFloat32		**Shape: (frequencyALength, frequencyAWidth)**
Description: Focused SLC image (RV)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image RV
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyA/numberOfSubSwaths		
Type: UByte		**Shape: scalar**
Description: Number of swaths of continuous imagery, due to transmit gaps		
	units	1
/science/LSAR/GSLC/grids/frequencyB/listOfPolarizations		
Type: string		**Shape: (numberOfFrequencyBPolarizations)**
Description: List of processed polarization layers with frequency B		
/science/LSAR/GSLC/grids/frequencyB/yCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive lines		
	long_name	Y coordinates spacing
	units	meters
/science/LSAR/GSLC/grids/frequencyB/xCoordinateSpacing		
Type: Float64		**Shape: scalar**
Description: Nominal spacing in meters between consecutive pixels		
	long_name	X coordinates spacing
	units	meters
/science/LSAR/GSLC/grids/frequencyB/rangeBandwidth		
Type: Float64		**Shape: scalar**
Description: Processed range bandwidth in hertz		
	units	hertz
/science/LSAR/GSLC/grids/frequencyB/azimuthBandwidth		
Type: Float64		**Shape: scalar**
Description: Processed azimuth bandwidth in hertz		
	units	hertz
/science/LSAR/GSLC/grids/frequencyB/centerFrequency		

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Type: Float64		**Shape: scalar**
Description: Center frequency of the processed image in hertz		
	units	hertz
/science/LSAR/GSLC/grids/frequencyB/slantRangeSpacing		
Type: Float64		**Shape: scalar**
Description: Slant range spacing of grid. Same as difference between consecutive samples in slantRange array		
	units	meters
/science/LSAR/GSLC/grids/frequencyB/zeroDopplerTimeSpacing		
Type: Float64		**Shape: scalar**
Description: Time interval in the along-track direction for raster layers. This is same as the spacing between consecutive entries in the zeroDopplerTime array		
	units	seconds
/science/LSAR/GSLC/grids/frequencyB/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/grids/frequencyB/xCoordinates		
Type: Float64		**Shape: (frequencyBWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/grids/frequencyB/yCoordinates		
Type: Float64		**Shape: (frequencyBLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/grids/frequencyB/mask		
Type: UByte		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Mask indicating the subswath number representing valid GSLC samples. Each GSLC pixel is assumed valid if all the pixels in the interpolation window are fully focused in the input RSLC. A value of `0` indicates that at least one RSLC pixel in the interpolation window is partially focused or invalid. Pixels outside of the radar acquisition extent are filled with the value `255`.		
	grid_mapping	projection
	long_name	Valid samples subswath mask

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	valid_min	0
	_FillValue	255
/science/LSAR/GSLC/grids/frequencyB/HH		
Type: CFloat32		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Focused SLC image (HH)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image HH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyB/HV		
Type: CFloat32		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Focused SLC image (HV)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image HV
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyB/VH		
Type: CFloat32		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Focused SLC image (VH)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image VH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyB/VV		
Type: CFloat32		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Focused SLC image (VV)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points

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	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image VV
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyB/RH		
Type: CFloat32		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Focused SLC image (RH)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image RH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyB/RV		
Type: CFloat32		**Shape: (frequencyBLength, frequencyBWidth)**
Description: Focused SLC image (RV)		
	min_imag_value	Minimum value of the imaginary part of the numeric data points
	max_imag_value	Maximum value of the imaginary part of the numeric data points
	mean_imag_value	Arithmetic average of the imaginary part of the numeric data points
	sample_stddev_imag	Standard deviation of the imaginary part of the numeric data points
	min_real_value	Minimum value of the real part of the numeric data points
	max_real_value	Maximum value of the real part of the numeric data points
	mean_real_value	Arithmetic average of the real part of the numeric data points
	sample_stddev_real	Standard deviation of the real part of the numeric data points
	long_name	Geocoded single-look complex image RVH
	grid_mapping	projection
	_FillValue	(nan+nan*j)
	units	1
/science/LSAR/GSLC/grids/frequencyB/numberOfSubSwaths		
Type: UByte		**Shape: scalar**
Description: Number of swaths of continuous imagery, due to transmit gaps		
	units	1

This document has been reviewed and determined not to contain export controlled technical data.

5.4 Calibration Information

Table 5-4 NISAR HDF5 variables related to calibration

Calibration-related variables		
/science/LSAR/GSLC/metadata/calibrationInformation/geometry/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/calibrationInformation/geometry/yCoordinates		
Type: Float64		**Shape: (calibrationLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/geometry/xCoordinates		
Type: Float64		**Shape: (calibrationWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/geometry/beta0		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: 2D LUT to convert DN to beta 0 assuming as a function of geographical location		
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/geometry/sigma0		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: 2D LUT to convert DN to sigma 0 assuming as a function of geographical location		
	_FillValue	nan
	grid_mapping	projection

This document has been reviewed and determined not to contain export controlled technical data.

	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/geometry/gamma0		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: 2D LUT to convert DN to gamma 0 as a function of geographical location		
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/yCoordinates		
Type: Float64		**Shape: (calibrationLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/xCoordinates		
Type: Float64		**Shape: (calibrationWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/HH		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/HV		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/VH		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/VV		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/RH		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/elevationAntennaPattern/RV		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/yCoordinates		
Type: Float64		**Shape: (calibrationLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/xCoordinates		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64		**Shape: (calibrationWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/HH		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/HV		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/VH		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/VV		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/RH		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/elevationAntennaPattern/RV		
Type: CFloat32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Complex two-way elevation antenna pattern		
	_FillValue	(nan+nan*j)
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure

This document has been reviewed and determined not to contain export controlled technical data.

	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/yCoordinates		
Type: Float64	**Shape: (calibrationLength)**	
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/xCoordinates		
Type: Float64	**Shape: (calibrationWidth)**	
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/HH		
Type: Float32	**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**	
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/HV		
Type: Float32	**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**	
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/VH		
Type: Float32	**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**	
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/VV		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/RH		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/noiseEquivalentBackscatter/RV		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/projection		
Type: UInt32		**Shape: scalar**
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/yCoordinates		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64		**Shape: (calibrationLength)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/xCoordinates		
Type: Float64		**Shape: (calibrationWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/HH		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/HV		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/VH		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/VV		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/RH		
Type: Float32		**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**

This document has been reviewed and determined not to contain export controlled technical data.

Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/noiseEquivalentBackscatter/RV		
Type: Float32	**Shape: (calibrationTimeLength, calibrationSlantRangeWidth)**	
Description: Noise equivalent backscatter in linear scale (units of DN^2)		
	mean_value	Arithmetic average of the numeric data points
	min_value	Minimum value of the numeric data points
	max_value	Maximum value of the numeric data points
	sample_stddev	Standard deviation of the numeric data points
	_FillValue	nan
	grid_mapping	projection
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/crosstalk/txHorizontalCrosspol		
Type: CFloat32	**Shape: ()**	
Description: Crosstalk in H-transmit channel expressed as ratio txV / txH		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/crosstalk/txVerticalCrosspol		
Type: CFloat32	**Shape: ()**	
Description: Crosstalk in V-transmit channel expressed as ratio txH / txV		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/crosstalk/rxHorizontalCrosspol		
Type: CFloat32	**Shape: ()**	
Description: Crosstalk in H-receive channel expressed as ratio rxV / rxH		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/crosstalk/rxVerticalCrosspol		
Type: CFloat32	**Shape: ()**	
Description: Crosstalk in V-receive channel expressed as ratio rxH / rxV		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/commonDelay		
Type: Float64	**Shape: scalar**	
Description: Range delay correction applied to all polarimetric channels		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/faradayRotation		
Type: Float64	**Shape: scalar**	
Description: Faraday rotation correction applied in processing		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HH/rfiLikelihood		
Type: Float64	**Shape: scalar**	
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HV/rfiLikelihood		
Type: Float64	**Shape: scalar**	
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VH/rfiLikelihood		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VV/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RH/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RV/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HH/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HV/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VH/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VV/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RH/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RV/rfiLikelihood		
Type: Float64		**Shape: scalar**
Description: Severity of radio frequency interference (RFI) contamination in the data. Value is in the interval [0,1], where 0: lowest severity, and 1: highest severity (or NaN if RFI detection was skipped)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HH/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to HH channel		
	units	meters

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HH/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to HH channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HH/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to HH channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HH/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to HH channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HV/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to HV channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HV/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to HV channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HV/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to HV channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/HV/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to HV channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VH/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to VH channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VH/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to VH channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VH/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to VH channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VH/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to VH channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VV/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to VV channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VV/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to VV channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VV/scaleFactor		

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Type: Float64		**Shape: scalar**
Description: Scale factor applied to VV channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/VV/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to VV channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RH/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to RH channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RH/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to RH channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RH/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to RH channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RH/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to RH channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RV/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to RV channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RV/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to RV channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RV/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to RV channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyA/RV/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to RV channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/commonDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to all polarimetric channels		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/faradayRotation		
Type: Float64		**Shape: scalar**
Description: Faraday rotation correction applied in processing		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HH/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to HH channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HH/differentialPhase		
Type: Float64		**Shape: scalar**

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Description: Phase correction applied to HH channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HH/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to HH channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HH/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to HH channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HV/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to HV channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HV/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to HV channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HV/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to HV channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/HV/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to HV channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VH/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to VH channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VH/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to VH channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VH/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to VH channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VH/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to VH channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VV/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to VV channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VV/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to VV channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VV/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to VV channel complex amplitude (at antenna boresite)		

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	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/VV/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to VV channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RH/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to RH channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RH/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to RH channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RH/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to RH channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RH/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to RH channel complex amplitude with respect to elevation angle		
	units	radians^-1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RV/differentialDelay		
Type: Float64		**Shape: scalar**
Description: Range delay correction applied to RV channel		
	units	meters
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RV/differentialPhase		
Type: Float64		**Shape: scalar**
Description: Phase correction applied to RV channel		
	units	radians
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RV/scaleFactor		
Type: Float64		**Shape: scalar**
Description: Scale factor applied to RV channel complex amplitude (at antenna boresite)		
	units	1
/science/LSAR/GSLC/metadata/calibrationInformation/frequencyB/RV/scaleFactorSlope		
Type: Float64		**Shape: scalar**
Description: Slope of scale factor applied to RV channel complex amplitude with respect to elevation angle		
	units	radians^-1

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5.5 Processing Information

Table 5-5 NISAR HDF5 variables related to processing parameters

Processing-related variables		
/science/LSAR/GSLC/metadata/processingInformation/parameters/azimuthChirpWeighting		
Type: Float32	**Shape: (chirpFFTFrequency)**	
Description: 1-D array in frequency domain for azimuth processing. This is used for processing L0b to L1. FFT length=256 (assumed)		
	units	1
/science/LSAR/GSLC/metadata/processingInformation/parameters/rangeChirpWeighting		
Type: Float32	**Shape: (chirpFFTFrequency)**	
Description: 1-D array in frequency domain for range processing. This is used for processing L0b to L1. FFT length=256 (assumed)		
	window_shape	Parameter that controls the shape of the apodization window
	window_name	Name of the apodization window, e.g., "Kaiser"
	units	1
/science/LSAR/GSLC/metadata/processingInformation/parameters/dryTroposphericGeolocationCorrectionApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the dry tropospheric correction has been applied to improve geolocation		
/science/LSAR/GSLC/metadata/processingInformation/parameters/wetTroposphericGeolocationCorrectionApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the wet tropospheric correction has been applied to improve geolocation		
/science/LSAR/GSLC/metadata/processingInformation/parameters/rangeIonosphericGeolocationCorrectionApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the range ionospheric correction has been applied to improve geolocation		
/science/LSAR/GSLC/metadata/processingInformation/parameters/azimuthIonosphericGeolocationCorrectionApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the azimuth ionospheric correction has been applied to improve geolocation		
/science/LSAR/GSLC/metadata/processingInformation/parameters/rfiMitigationApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if an RFI mitigation has been applied		
/science/LSAR/GSLC/metadata/processingInformation/parameters/ellipsoidalFlatteningApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the GSLC phase has been flattened with respect to a zero height ellipsoid		
/science/LSAR/GSLC/metadata/processingInformation/parameters/topographicFlatteningApplied		
Type: string	**Shape: scalar**	
Description: Flag to indicate if the GSLC phase has been flattened with respect to topographic height using a DEM		
/science/LSAR/GSLC/metadata/processingInformation/parameters/geocoding/snapToGridX		
Type: Float64	**Shape: scalar**	
Description: Snap to the grid along the X direction with the origin set at 0. The X-coordinate of the scene's upper-left corner is a multiple of this value		
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/geocoding/snapToGridY		
Type: Float64	**Shape: scalar**	
Description: Snap to the grid along the Y direction with the origin set at 0. The Y-coordinate of the scene's upper-left corner is a multiple of this value		
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/referenceTerrainHeight		
Type: Float32	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**	

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Description: Reference Terrain Height as a function of geographical location		
	_FillValue	nan
	grid_mapping	projection
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/projection		
Type: UInt32	**Shape: scalar**	
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/processingInformation/parameters/yCoordinates		
Type: Float64	**Shape: (dopplerCentroidLength)**	
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/xCoordinates		
Type: Float64	**Shape: (dopplerCentroidWidth)**	
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyA/projection		
Type: UInt32	**Shape: scalar**	
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.

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	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyA/yCoordinates		
Type: Float64	**Shape: (dopplerCentroidLength)**	
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyA/xCoordinates		
Type: Float64	**Shape: (dopplerCentroidWidth)**	
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyA/dopplerCentroid		
Type: Float64	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**	
Description: 2D LUT of Doppler centroid for frequency A		
	_FillValue	nan
	grid_mapping	projection
	units	hertz
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyB/projection		
Type: UInt32	**Shape: scalar**	
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyB/yCoordinates		
Type: Float64	**Shape: (dopplerCentroidLength)**	
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters

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/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyB/xCoordinates		
Type: Float64	**Shape: (dopplerCentroidWidth)**	
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/parameters/frequencyB/dopplerCentroid		
Type: Float64	**Shape: (dopplerCentroidLength, dopplerCentroidWidth)**	
Description: 2D LUT of Doppler centroid for frequency B		
	_FillValue	nan
	grid_mapping	projection
	units	hertz
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/azimuthIonosphere		
Type: Float64	**Shape: (timingCorrectionLength, timingCorrectionWidth)**	
Description: 2D lookup table of azimuth ionosphere timing correction derived from Total Electron Content data		
	_FillValue	nan
	units	seconds
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/slantRangeIonosphere		
Type: Float64	**Shape: (timingCorrectionLength, timingCorrectionWidth)**	
Description: 2D lookup table of line-of-sight ionosphere timing correction derived from Total Electron Content data		
	_FillValue	nan
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/slantRangeSolidEarthTides		
Type: Float64	**Shape: (timingCorrectionLength, timingCorrectionWidth)**	
Description: 2D lookup table of line-of-sight solid Earth tides timing correction		
	_FillValue	nan
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/slantRange		
Type: Float64	**Shape: (timingCorrectionWidth)**	
Description: Slant range dimension corresponding to the timing correction lookup tables		
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/slantRangeSpacing		
Type: Float64	**Shape: scalar**	
Description: Slant range spacing of the timing correction lookup tables		
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/zeroDopplerTime		
Type: Float64	**Shape: (timingCorrectionLength)**	
Description: Zero Doppler time dimension since UTC epoch corresponding to the timing correction lookup tables		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyA/zeroDopplerTimeSpacing		
Type: Float64	**Shape: scalar**	
Description: Time interval in the along-track direction of the timing correction lookup tables		
	units	seconds
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/azimuthIonosphere		
Type: Float64	**Shape: (timingCorrectionLength, timingCorrectionWidth)**	
Description: 2D lookup table of azimuth ionosphere timing correction derived from Total Electron Content data		
	_FillValue	nan
	units	seconds
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/slantRangeIonosphere		
Type: Float64	**Shape: (timingCorrectionLength, timingCorrectionWidth)**	
Description: 2D lookup table of line-of-sight ionosphere timing correction derived from Total Electron Content data		
	_FillValue	nan
	units	meters

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/slantRangeSolidEarthTides		
Type: Float64	**Shape: (timingCorrectionLength, timingCorrectionWidth)**	
Description: 2D lookup table of line-of-sight solid Earth tides timing correction		
	_FillValue	nan
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/slantRange		
Type: Float64	**Shape: (timingCorrectionWidth)**	
Description: Slant range dimension corresponding to the timing correction lookup tables		
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/slantRangeSpacing		
Type: Float64	**Shape: scalar**	
Description: Slant range spacing of the timing correction lookup tables		
	units	meters
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/zeroDopplerTime		
Type: Float64	**Shape: (timingCorrectionLength)**	
Description: Zero Doppler time dimension since UTC epoch corresponding to the timing correction lookup tables		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GSLC/metadata/processingInformation/timingCorrections/frequencyB/zeroDopplerTimeSpacing		
Type: Float64	**Shape: scalar**	
Description: Time interval in the along-track direction of the timing correction lookup tables		
	units	seconds
/science/LSAR/GSLC/metadata/processingInformation/parameters/runConfigurationContents		
Type: string	**Shape: scalar**	
Description: Contents of the run configuration file with parameters used for processing		
/science/LSAR/GSLC/metadata/processingInformation/algorithms/softwareVersion		
Type: string	**Shape: scalar**	
Description: Software version used for processing		
/science/LSAR/GSLC/metadata/processingInformation/algorithms/demInterpolation		
Type: string	**Shape: scalar**	
Description: DEM interpolation method		
/science/LSAR/GSLC/metadata/processingInformation/algorithms/geocoding		
Type: string	**Shape: scalar**	
Description: Geocoding algorithm		
/science/LSAR/GSLC/metadata/processingInformation/inputs/l1SlcGranules		
Type: string	**Shape: (numberOfInputL0BFiles)**	
Description: List of input L1 products used		
/science/LSAR/GSLC/metadata/processingInformation/inputs/orbitFiles		
Type: string	**Shape: (numberOfInputOrbitFiles)**	
Description: List of input orbit files used		
/science/LSAR/GSLC/metadata/processingInformation/inputs/tecFiles		
Type: string	**Shape: (numberOfInputTecFiles)**	
Description: List of input total electron content (TEC) files used		
/science/LSAR/GSLC/metadata/processingInformation/inputs/configFiles		
Type: string	**Shape: (numberOfInputConfigFiles)**	
Description: List of input config files used		
/science/LSAR/GSLC/metadata/processingInformation/inputs/demSource		
Type: string	**Shape: scalar**	
Description: Description of the input digital elevation model (DEM)		

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5.6 Other Radar Metadata

Table 5-6 NISAR HDF5 variables related to radar metadata

Radar metadata-related variables		
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/boundingBox		
Type: string	**Shape: scalar**	
Description: OGR compatible WKT representing the bounding box of the product (the union of the geocoded grid for available frequency A and frequency B image datasets). Horizontal coordinates are X and Y directions of the geocoded grid coordinate system, and the vertical coordinate is the height above the WGS84 ellipsoid in meters. The first point corresponds to the top left edge of the bounding box. The remaining points are traversed in counterclockwise order on the map. The fifth point closes the bounding box by repeating the first point		
	epsg	EPSG code
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/staticLayersDataAccess		
Type: string	**Shape: scalar**	
Description: Location of the static layers product associated with this product (URL or DOI)		
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/ceosAnalysisReadyDataProductType		
Type: string	**Shape: scalar**	
Description: CEOS Analysis Ready Data (CARD) product type		
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/ceosAnalysisReadyDataDocumentIdentifier		
Type: string	**Shape: scalar**	
Description: CEOS Analysis Ready Data (CARD) document identifier		
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/geometricAccuracy/bias/y		
Type: Float32	**Shape: scalar**	
Description: An estimate of the localization error bias in the Y/northing direction		
	epsg	EPSG code
	units	meters
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/geometricAccuracy/bias/x		
Type: Float32	**Shape: scalar**	
Description: An estimate of the localization error bias in the X/easting direction		
	epsg	EPSG code
	units	meters
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/geometricAccuracy/standardDeviation/y		
Type: Float32	**Shape: scalar**	
Description: An estimate of the localization error standard deviation in the Y/northing direction		
	epsg	EPSG code
	units	meters
/science/LSAR/GSLC/metadata/ceosAnalysisReadyData/geometricAccuracy/standardDeviation/x		
Type: Float32	**Shape: scalar**	
Description: An estimate of the localization error standard deviation in the X/easting direction		
	epsg	EPSG code
	units	meters
/science/LSAR/GSLC/metadata/orbit/time		
Type: Float64	**Shape: (orbitListLength)**	
Description: Time vector record. This record contains the time since UTC epoch corresponding to position and velocity records		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GSLC/metadata/orbit/position		
Type: Float64	**Shape: (orbitListLength, tripletxyz)**	

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Description: Position vector record. This record contains the platform position data with respect to WGS84 G1762 reference frame		
	units	meters
/science/LSAR/GSLC/metadata/orbit/velocity		
Type: Float64	**Shape: (orbitListLength, tripletxyz)**	
Description: Velocity vector record. This record contains the platform velocity data with respect to WGS84 G1762 reference frame		
	units	meters / second
/science/LSAR/GSLC/metadata/orbit/orbitType		
Type: string	**Shape: scalar**	
Description: Orbit product type, either "FOE", "NOE", "MOE", "POE", or "Custom", where "FOE" stands for Forecast Orbit Ephemeris, "NOE" is Near real-time Orbit Ephemeris, "MOE" is Medium precision Orbit Ephemeris, and "POE" is Precise Orbit Ephemeris		
/science/LSAR/GSLC/metadata/orbit/interpMethod		
Type: string	**Shape: scalar**	
Description: Orbit interpolation method, either "Hermite" or "Legendre"		
/science/LSAR/GSLC/metadata/attitude/time		
Type: Float64	**Shape: (attitudeListLength)**	
Description: Time vector record. This record contains the time since UTC epoch corresponding to attitude and quaternion records		
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GSLC/metadata/attitude/quaternions		
Type: Float64	**Shape: (attitudeListLength, quaternions)**	
Description: Attitude quaternions (q0, q1, q2, q3)		
	units	1
/science/LSAR/GSLC/metadata/attitude/eulerAngles		
Type: Float64	**Shape: (attitudeListLength, tripletxyz)**	
Description: Attitude Euler angles (roll, pitch, yaw)		
	units	degrees
/science/LSAR/GSLC/metadata/attitude/attitudeType		
Type: string	**Shape: scalar**	
Description: Attitude type, either "FRP", "NRP", "PRP, or "Custom", where "FRP" stands for Forecast Radar Pointing, "NRP" is Near Real-time Pointing, and "PRP" is Precise Radar Pointing		

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5.7 Radar Grid

Table 5-7 NISAR HDF5 variables related to radar grid

Metadata cube-related variables		
/science/LSAR/GSLC/metadata/radarGrid/zeroDopplerAzimuthTime		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Zero Doppler azimuth time in seconds since UTC epoch		
	_FillValue	nan
	long_name	Zero Doppler azimuth time
	grid_mapping	projection
	units	seconds since YYYY-mm-ddTHH:MM:SS
/science/LSAR/GSLC/metadata/radarGrid/slantRange		
Type: Float64	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Slant range in meters		
	_FillValue	nan
	long_name	Slant range
	grid_mapping	projection
	units	meters
/science/LSAR/GSLC/metadata/radarGrid/incidenceAngle		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Incidence angle is defined as the angle between the LOS vector and the normal to the ellipsoid at the target height		
	valid_max	90.0
	valid_min	0.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Incidence angle
	units	degrees
/science/LSAR/GSLC/metadata/radarGrid/losUnitVectorX		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: East component of unit vector of LOS from target to sensor		
	valid_max	1.0
	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	LOS unit vector X
	units	1
/science/LSAR/GSLC/metadata/radarGrid/losUnitVectorY		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: North component of unit vector of LOS from target to sensor		
	valid_max	1.0
	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	LOS unit vector Y
	units	1
/science/LSAR/GSLC/metadata/radarGrid/alongTrackUnitVectorX		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: East component of unit vector along ground track		
	valid_max	1.0

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	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Along-track unit vector X
	units	1
/science/LSAR/GSLC/metadata/radarGrid/alongTrackUnitVectorY		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: North component of unit vector along ground track		
	valid_max	1.0
	valid_min	-1.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Along-track unit vector Y
	units	1
/science/LSAR/GSLC/metadata/radarGrid/elevationAngle		
Type: Float32	**Shape: (radarCubeHeight, radarCubeLength, radarCubeWidth)**	
Description: Elevation angle is defined as the angle between the LOS vector and the normal to the ellipsoid at the sensor		
	valid_max	90.0
	valid_min	0.0
	_FillValue	nan
	grid_mapping	projection
	long_name	Elevation angle
	units	degrees
/science/LSAR/GSLC/metadata/radarGrid/groundTrackVelocity		
Type: Float64	**Shape: (radarCubeLength, radarCubeWidth)**	
Description: Absolute value of the platform velocity scaled at the target height		
	_FillValue	nan
	grid_mapping	projection
	long_name	Ground track velocity
	units	meters / second
/science/LSAR/GSLC/metadata/radarGrid/projection		
Type: UInt32	**Shape: scalar**	
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	ellipsoid	Projection ellipsoid
	epsg_code	Projection EPSG code
	false_easting	The value added to all abscissa values in the rectangular coordinates for a map projection.
	false_northing	The value added to all ordinate values in the rectangular coordinates for a map projection.
	grid_mapping_name	Grid mapping variable name
	inverse_flattening	Inverse flattening of the ellipsoidal figure
	latitude_of_projection_origin	The latitude chosen as the origin of rectangular coordinates for a map projection.
	longitude_of_projection_origin	The longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum.
	longitude_of_central_meridian	The line of longitude at the center of a map projection generally used as the basis for constructing the projection.
	scale_factor_at_central_meridian	A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance along the central meridian.
	semi_major_axis	Semi-major axis
	spatial_ref	Spatial reference
	utm_zone_number	UTM zone number
/science/LSAR/GSLC/metadata/radarGrid/xCoordinates		

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Type: Float64		**Shape: (radarCubeWidth)**
Description: X coordinates in specified projection		
	long_name	X coordinates of projection
	standard_name	projection_x_coordinate
	units	meters
/science/LSAR/GSLC/metadata/radarGrid/yCoordinates		
Type: Float64		**Shape: (radarCubeWidth)**
Description: Y coordinates in specified projection		
	long_name	Y coordinates of projection
	standard_name	projection_y_coordinate
	units	meters
/science/LSAR/GSLC/metadata/radarGrid/heightAboveEllipsoid		
Type: Float64		**Shape: (radarCubeHeight)**
Description: Height values above WGS84 Ellipsoid corresponding to the radar grid		
	standard_name	height_above_reference_ellipsoid
	units	meters

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6 METADATA CUBE

In this section, we provide an overview of the metadata cubes used to store spatially-varying ancillary data in the secondary layers of the NISAR L-SAR product HDF5 granules. Metadata cubes are represented as three-dimensional arrays in the NISAR product HDF5 modules (Figure 6-1). The axes of the array are interpreted as (height, increasing azimuth time, and increasing slant range) in case of radar geometry products and as (height, decreasing northing, and increasing easting) in case of geocoded products. The data is organized with height as the first axis. Each height layer is the same size. Metadata cubes will have fixed grid spacing and will allow for easy merging when multiple products along the same imaging track are to be concatenated. The metadata fields on this coarse resolution grid will be evaluated using traditional radar processing approaches without approximations. The metadata cube will also span a field slightly larger than the original image product to allow users to interpolate data without introducing edge effects. Such low-resolution representation of slowly varying parameters has been demonstrated for InSAR products and processing [RD7].

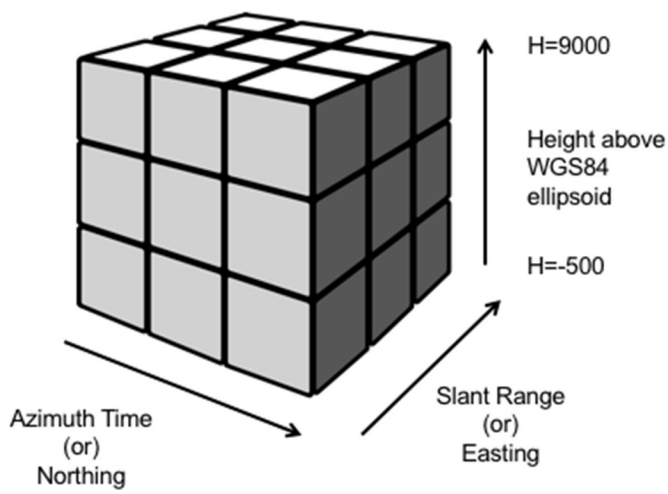

Figure 6-1. Metadata cube layer schematic

6.1 Metadata Cube Interpolation Example

We provide here a conceptual example of how these metadata cubes can be used. Let us consider a GSLC product on a UTM Zone 10 grid (Table 6-1). We use a geocoded product for the demonstration but the presented approach can be easily extended to radar coordinate products by replacing northing axis by azimuth time and easting axis by slant range.

Table 6-1. Example metadata cube properties

Name	Value	Description
Primary layer properties		
xmin	100000.0	Easting of the first column (m)

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Name	Value	Description
xmax	340000.0	Easting of the last column (m)
dx	30.0	Column spacing in Easting (m)
Nx	8001	Number of columns
ymax	570000.0	Northing of first row (m)
ymin	330000.0	Northing of last row (m)
dy	-30.0	Row spacing in Northing (m). Negative to emphasize North-up imagery in geocoded products
Ny	8001	Number of rows
Metadata cube properties		
Cxmin	97000.0	Easting of first column (m)
Cxmax	343000.0	Easting of last column (m)
Cdx	1000.0	Column spacing in Easting (m)
CNx	247	Number of columns
Cymax	579000.0	Northing of first row (m)
Cymin	321000.0	Northing of last row(m)
Cdy	-3000.0	Row spacing in Northing (m). Negative to emphasize North-up imagery in geocoded products
CNy	87	Number of rows
Czmin	-1500	Height of the first layer (m)
Czmax	9000	Height of the last layer (m)
Cdz	1500	Layer spacing in height (m)
CNz	8	Number of height layers

Suppose we are interested in computing the Perpendicular Baseline (Bperp) at a pixel of interest located at UTM coordinates point (Px,Py). Since these are coordinates on a map domain, we can look up a DEM to get the height at this point. The three-dimensional point of interest then becomes (Px, Py, h(Px,Py)).

The metadata cube for Perpendicular baseline can be thought of as a three-dimensional field Bperp(x,y,z) – even though it is oriented as (Nz,Ny,Nx) in the HDF5 file for ease of use with a GIS. The user can use standard built-in regular grid three-dimensional interpolation routines in languages like MATLAB (e.g, interp3), IDL or Python (e.g, RegularGridInterpolator) to interpolate the Bperp array. We recommend cubic interpolation for best results. If a three-dimensional interpolator is not available, one could use two-dimensional cubic interpolation for each height layer followed by a one-dimensional cubic interpolation in the following manner:

1. Populate f(i), i=0,…Nz-1 by two-dimensional cubic interpolation of each height layer:

$$f(i) = Bperp\left[i, \frac{Py - Cymax}{Cdy}, \frac{Px - Cxmin}{Cdx}\right]$$

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where the numbers in the square brackets indicate indices into the three-dimensional cube. For example, if we are interested in the point (107590.0 East, 555870.0 North, 300.0 Height), we would interpolate at Row 7.71 and Column 10.59 for each height layer.

2. Interpolate f(i) using one-dimensional cubic interpolation:

$$Bperp\big(Px, Py, h(Px, Py)\big) = f\left[\frac{h(Px, Py) - Czmin}{Cdz}\right]$$

where the number in the square bracket indicates an index into a one-dimensional array. For example, for a height value of 200.0, we would interpolate at an index of 1.2.

6.2 Metadata Cube Usage Note

Note that the metadata cubes are designed to accommodate one double-precision cube within 1 MB of memory, allowing for information to be easily stored in memory for on-the-fly computation within GIS frameworks or software without much overhead. The metadata cubes are not a replacement for traditional SAR processing approaches or very high-resolution analyses. They are meant to facilitate rapid processing and analysis by non-experts and will serve the needs for most SAR applications. Analyses show that the geolocation error is on the order of 1.5 cm due to interpolation which is significantly smaller than errors from sources such as DEM, orbits, and atmospheric path delay. Interpolation errors for each of the metadata layers will be reported after additional study.

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APPENDIX A: ACRONYMS

ADT	Algorithm Development Team
ANF	Area Normalization Factor
AT	Along Track
ATBD	Algorithm Theoretical Basis Document
AWS	Amazon Web Services
BFPQ	Block (adaptive) Floating-Point Quantization (adaptive may indicate implementation options)
Cal/Val	Calibration and Validation (also sometimes cal/val)
CDR	Critical Design Review
CF	Climate and Forecast
CPU	Central Processing Unit
CRSD	Calibration Raw Signal Data
CSV	Comma-separated values
DAAC	Distributed Active Archive Center
DBF	Digital Beam Forming
DEM	Digital Elevation Model
DM	Diagnostic Mode
DN	Digital Number
EAR	Export Administration Regulations
EASE	Equal-Area Scalable Earth
ECMWF	European Centre for Medium-Range Weather Forecasts
ECEF	Earth Centered Earth Fixed
ER#.#	Engineering Release #.#
ERA5	ECMWF Reanalysis 5th generation
FFT	Fast Fourier Transform
FM	Frequency Modulation
FOE	Forecast Orbit Ephemeris
FOV	Field of View
GCOV	Geocoded Polarimetric Covariance (GCOV)
GCP	Ground Control Point
GDAL	Geospatial Data Abstraction Library
GDS	Ground Data System
GeoTIFF	Geographic Tagged Image File Format
GIS	Geographic Information System
GMTED	Global Multi-resolution Terrain Elevation Data
GNSS	Global Navigation Satellite System
GOFF	Geocoded Pixel Offsets (GOFF)
GPU	Graphics Processing Unit
GSLC	Geocoded Single Look Complex (GSLC)
GUNW	Geocoded Unwrapped Interferogram (GUNW)
HH	Horizontal-transmit, Horizontal-receive polarization
HK, HKTM	Housekeeping Telemetry

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HDF5	Hierarchical Data Format version 5
HV	Horizontal-transmit, Vertical-receive polarization
ICU	Integrated Correlation Unit
InSAR	Interferometric Synthetic Aperture Radar
ISCE	InSAR Scientific Computing Environment
ISCE3	InSAR Scientific Computing Environment Enhanced Edition (for NISAR)
ISO	International Organization for Standardization
ISRO	Indian Space Research Organisation (British spelling)
JPL	Jet Propulsion Laboratory
JSON	JavaScript Notation
L0B	Level-0B (data)
L1	Level-1 (data)
L2	Level-2 (data)
L3	Level-3 (data)
LRR	[JPL] Limited Release Request
LRS	[JPL] Limited Release System
LUT	Lookup Table
Mbps	Megabits per second
MHz	Megahertz
MOE	Medium-precision Orbit Ephemeris
NASA	National Aeronautics and Space Administration
NETCDF4	Network Common Data Format 4 (also netCDF4)
NISAR	NASA-ISRO Synthetic Aperture Radar
NOE	Near-Realtime Orbit Ephemeris
OpenMP	Open Multi-Processing
PCM	Process Control Management
PDF	Portable Document Format (often pdf)
PDR	Preliminary Design Review
POD	Precision Orbit Determination
POE	Precision Orbit Ephemeris
PRF	Pulse Repetition Frequency
QA	Quality Assurance
R#.#	Release #.# (.0 often not used)
REE	Radar Echo Emulator
RFI	Radio Frequency Interference
RIFG	Range-Doppler Interferogram (RIFG)
ROFF	Range-Doppler Pixel Offsets (ROFF)
RRSD	Raw Radar Signal Data
RRST	Raw Radar Signal Telemetry
RSLC	Range-Doppler Single Look Complex (RSLC)
RTC	Radiometric Terrain Correction
RUNW	Range-Doppler UnWrapped Interferogram (RUNW)
RV	Right-circular, V-receive compact polarization

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SAR	Synthetic Aperture Radar (L-SAR: L-band. S-SAR: S-band)
SAS	Science Algorithm Software
SDS	Science Data System
SDT	Science Definition Team
SIS	Software Interface Specification
SLC	Single Look Complex
SME2	Soil Moisture product based on a 200-meter global EASE Grid projection
SMAP	Soil Moisture Active Passive (Mission)
SNAPHU	Statistical-cost, Network-flow Algorithm for Phase Unwrapping
SRTM	Shuttle Radar Topography Mission
ST	Science Team
SWST	Sampling Window Start Time
TAI	International Atomic Time (Temps Atomique International)
TCF	Terrain Correction Factor
TEC	Total Electron Content
TFdb	Trackframe Database
SWST	Sampling Window Start Time
UR	Urgent Response
UTC	Universal Time Coordinated
UTM	Universal Transverse Mercator
VH	Vertical-transmit, Horizontal-receive polarization
VV	Vertical-transmit, Vertical-receive polarization
WGS84	World Geodetic System 84
XML	eXtensible Markup Language (xml in code)
YAML	YAML Ain’t Markup Language

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APPENDIX B: GEOCODED PRODUCT GRIDS

NISAR L2 products will be generated on a pre-defined Track/Frame system. The projection system for a particular frame will be available to the users as a predefined map and will be held constant through the life of the mission. Each L2 HDF5 granule itself will include information indicating the projection used for the product.

Map Projections

The NISAR SDS is able to ingest any Digital Elevation Model whose vertical datum represents height above the WGS84 Ellipsoid and the horizontal datum can be represented by an European Petroleum Standards Group (EPSG) code for generating geocoded product. Table 7-1 lists the various projection systems used to output L2 geocoded products.

Table B-1. Projection Systems for NISAR L2 Products

EPSG code	PROJ.4 string	Common Name	Geographical scope
3031	+proj=stere +lat_0=-90 +lat_ts=-71 +lon_0=0 +k=1 +x_0=0 +y_0=0 +datum=WGS84 +units=m +no_defs	Antarctic Polar Stereographic	Antarctica and Southern Hemisphere Sea Ice
3413	+proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +k=1 +x_0=0 +y_0=0 +datum=WGS84 +units=m +no_defs	NSIDC Sea Ice Polar Stereographic North	Greenland and Northern Hemisphere Sea Ice
32601-32660	+proj=utm +zone=X-32600 +datum=WGS84 +units=m +no_defs	UTM Zone North	Northern Hemisphere Land except Greenland
32701-32760	+proj=utm +zone=X-32700 +south +datum=WGS84 +units=m +no_defs	UTM Zone South	Southern Hemisphere Land except Antarctica

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